Instructor's Solutions Manual

Foundations of Sensitivity Analysis: Theory, Computation, and Applications

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Contents

How to use this manual. Solutions are organized by chapter and exercise number, matching the labeling in the main text. For each exercise, the Bloom's Taxonomy level is given in brackets. L2 solutions are concise; L5–L6 solutions provide model answers that can be adapted for grading rubrics. Where MATLAB verification is natural, it is included.

For exercises marked *Instructor note*, the note provides grading guidance, common student errors, or pedagogical context.

1 Chapter 1: Introduction and Overview

Exercise 1.1 *[Bloom L2]*

Solution.

A **parameter of interest** (POI) is an input to the model whose value may be uncertain or whose influence we wish to quantify. A **quantity of interest** (QOI) is an output of the model that we care about — the number we are trying to predict or understand.

For an antibiotic-resistance model in a hospital, two POIsmight be: the probability that a patient receives an antibiotic in a given hospital stay, and the probability that a resistant strain is transmitted per nurse-patient contact. Two QOIsmight be: the fraction of patients colonized with a resistant strain after 30 days, and the mean length of stay for infected patients.

Running the model tells us the QOIat the nominal parameter values. Sensitivity analysis tells us *how much the* QOI*would change* if a POIwere different — that is, which parameters matter most for the model's predictions, so we can focus data collection and intervention design on those parameters.

Instructor note: Accept any reasonable POI/QOIpair consistent with the antibiotic-resistance context. The key distinction: POIsare inputs controlled or measured by the investigator; QOIsare outputs that answer the policy or scientific question. Students sometimes confuse state variables with POIs; clarify that parameters govern the dynamics, while state variables evolve according to those parameters.

Exercise 1.2 *[Bloom L2]*

Solution.

Forward SA answers "What if?" questions: if this POIchanges, how does the QOIrespond? It computes sensitivity by varying each POIone at a time and observing the effect on the QOI. The cost scales with the number of POIs(one computation per POI).

Adjoint SA answers "How come?" questions: given that the QOIhas the value it does, which parameters are responsible? It computes sensitivity by working backward from the QOIto the POIs. The cost scales with the number of QOIs(one computation per QOI).

Classifying the four questions:

(a) *If the mean latent period increases by 10%, how does peak infection count change?* — **Forward** question (one POI, asking about effect on QOI).

(b) *The peak infection count exceeded the hospital capacity. Which parameters are most responsible?* — **Adjoint** question (one QOI, asking which POIsdrove it).

(c) *We have 20 uncertain parameters. Which 3 are worth measuring precisely?* — **Forward** (or global) SA: compute SIs for all 20 parameters and identify the largest.

(d) *We want $\partial J/\partial p$ for 50 parameters where J is a single regulatory threshold.* — **Adjoint**: one QOI, many POIs, so adjoint is efficient.

Instructor note: Part (d) is a good opportunity to preview the cost comparison (Chapter 5): adjoint gives all 50 sensitivities at the cost of one computation, vs. 50 computations for forward SA.

Exercise 1.3 *[Bloom L4]*

Solution.

(a) *Climate model, 500 parameters, identify 10 most influential on mean surface temperature, 8 hours per run.*

Morris screening (or adjoint, if available). With 8 hours per run, a full Sobol analysis at $N = 2000$ would cost $N(m + 2) = 1{,}004{,}000$ runs, or about 916 years of computation. Morris screening with $r = 10$ trajectories costs $r(m + 1) = 5010$ runs ($\approx$4 years) — still expensive, but feasible on a cluster. Adjoint SA would cost the equivalent of one forward run plus storage, giving all 500 sensitivities simultaneously. If the adjoint is available, use it; otherwise, Morris screening on a cluster is the practical alternative.

(b) *Pollution transport model, 6 parameters, threshold exceedance at one monitoring station.*

Adjoint SA. The QOIis a single threshold value at one location — exactly the scenario where the adjoint is optimal. With 6 parameters and 1 QOI, adjoint costs 1 backward solve vs. 6 forward perturbations. The adjoint gives the complete gradient of the threshold value with respect to all 6 parameters simultaneously.

(c) *Pharmacokinetic model, 8 parameters, patient outcome is AUC (area under the drug-concentration curve), nonlinear dose-response.*

Global SA (Sobol indices). The dose-response is nonlinear, so local SA (forward or adjoint) will miss nonlinear contributions and interactions. Sobol indices capture the full variance-based sensitivity across the parameter uncertainty range. With 8 parameters, a Sobol analysis at $N = 2000$ requires $2000 \times 10 = 20{,}000$ evaluations, which is typically feasible for a pharmacokinetic ODE model.

(d) *Structural engineering FEM model, 200 parameters, single safety factor* QOI*, model takes 10 seconds per run.*

Adjoint SA. Single QOI(safety factor), 200 parameters, and 10 seconds per run: Sobol at $N = 2000$ would take $2000 \times 202 = 404{,}000$ evaluations ≈ 47 days. Adjoint gives all 200 sensitivities in $\approx$20 seconds (one forward + one backward solve). The adjoint is the clear choice when the model is differentiable and $m \gg r$.

Instructor note: This exercise builds intuition for the decision table (Table 4.2 in the text). Common student errors: choosing global SA for (b) because "nonlinearity might be present" (the model is not stated to be nonlinear, and the threshold QOIis a single value at one station, so adjoint is correct); or choosing forward SA for (d) without computing the cost.

Exercise 1.4 *[Bloom L5]*

Solution.

Model answer (acceptable answers will vary in specifics but should include all four required components):

SA plan for a dengue fever model with 9 parameters.

QOI choice. I would use peak infectious count $\max_t I(t)$ as the primary QOI, because it determines whether the healthcare system is overwhelmed — the operationally critical threshold. A secondary QOIwould be the basic reproduction number R_0 (analytically tractable and directly related to whether an epidemic occurs). The relationship between the two provides a consistency check.

Identifying the 3 parameters to measure. Run a local SA using the forward sensitivity equations (analytic if the model is an ODE, or centered finite differences otherwise) to compute all 9 sensitivity indices $S_{p_j}^{\text{peak } I}$. Rank by $|S_{p_j}^{\cdot}|$. The three parameters with the largest absolute SI values are the priority for precise measurement. If two parameters appear in the model only as a product (structural correlation), treat the product as a single effective parameter.

Before reporting, also run a Morris screening (15 trajectories, cost: $15 \times 10 = 150$ model evaluations) to flag any parameters where nonlinearity makes the local SI misleading.

Additional information needed for intervention design. SA tells us which parameters *matter most for predictions*, not which can be controlled by public health interventions. I would cross-reference the top-ranked parameters with a table of interventions that affect each parameter (e.g., insecticide spraying affects vector contact rate; surveillance programs affect reporting rate). Only parameters that are both high-sensitivity *and* actionable with available resources should drive intervention recommendations.

Limitation. The most important limitation is that local SA is a linearization: the sensitivity indices are derivatives at the nominal parameter values. If the model is highly nonlinear (e.g., near a threshold between epidemic and no-epidemic regimes), local SA may not accurately reflect sensitivity across the full plausible parameter range. I would flag this to the team and recommend a global SA (Sobol) on the top 4–5 parameters identified by local SA, to check for nonlinear effects.

Instructor note: Grading rubric: 1 point for QOIjustification; 2 points for the parameter-identification procedure (must mention a specific SA method and be operationally specific about the ranking); 1 point for the intervention-actionability distinction; 1 point for identifying a genuine limitation of local SA. Deduct 1 point if student recommends global SA for all 9 parameters without noting the cost implication (dengue models are often ODEs; full Sobol is feasible, so this is not wrong, but the question asks them to use SA to guide *field* measurement decisions, which requires identifying the top 3 specifically).

Exercise 1.5 *[Bloom L6]*

Solution.

Problem 1: Conflation of "all parameters simultaneously" with global SA.

Computing local sensitivity indices (derivatives at best-fit values) does *not* analyze all parameters simultaneously. Local SA perturbs one parameter at a time, holding all others fixed. The statement implies that the joint effect of simultaneous parameter uncertainty was assessed, but it was not. What should have been done: either (a) state that local SA was used, which analyzes one parameter at a time and captures only linear, individual effects; or (b) if the goal was to assess robustness to joint uncertainty, perform global SA (e.g., Sobol indices), which does evaluate sensitivity across the full multi-dimensional parameter space.

Problem 2: "Robust to parameter uncertainty" is not a valid conclusion from local SI values.

Sensitivity indices measure the *rate of change* of the QOIwith respect to each parameter. A model with all small SIs at the best-fit values is locally insensitive near that point, but could be highly sensitive elsewhere (e.g., near a bifurcation). "Robust" should mean that the QOIchanges little across the full plausible range of parameter values — a global property, not a local one. What should have been done: specify a plausible parameter uncertainty range for each parameter, compute the resulting range of QOIvalues, and only then make a robustness claim.

Optional additional problem (award bonus credit): Local SA at "best-fit parameter values" is circular if the model was fitted to the same data being used to validate it. Sensitivities may be artificially small near the best-fit point if the fitting procedure has forced the model to be locally insensitive to parameter changes.

Instructor note: This is a Bloom L6 synthesis exercise. Full credit requires identifying problems that go beyond surface-level criticism. The strongest answers connect "local vs. global" to the mathematical definition of the SI (derivative vs. variance decomposition) rather than simply noting vague imprecision. Partial credit (1 point each) for correct identification of any two genuine methodological issues with specific alternative recommendations.

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2 Chapter 2: The Sensitivity Index

Exercise 2.1 *[Bloom L2]*

Solution.

The sign of a sensitivity index tells you the *direction* of the relationship: a positive SI means the QOIincreases when the POIincreases; a negative SI means the QOIdecreases when the POIincreases. The sign is the sign of the underlying derivative $\partial q/\partial p$, scaled by p/q (both positive under the interpretable-parameterization convention).

(a) *Lung capacity vs. cigarettes per day.* Lung capacity decreases as cigarette consumption increases (medical consensus). Therefore $\partial q/\partial p < 0$, so the SI is **negative**.

(b) *Crop yield vs. annual rainfall (near the optimum).* Crop yields typically increase with rainfall up to an optimum and then decrease (waterlogging). The sign depends on which side of the optimum we are evaluating. Near the dry end of the feasible range, yield increases with rainfall: **positive** SI. Near the wet end (waterlogging), yield decreases: **negative** SI. The question specifies "near a region of the parameter space"; the sign must be specified relative to that region, and the answer should note the non-monotonicity.

(c) *R_0 vs. γ (recovery rate) for SIR.* $R_0 = k\beta/\gamma$, so $\partial R_0/\partial\gamma < 0$: faster recovery reduces R_0. The SI is **negative**: $S_\gamma^{R_0} = -1$.

Instructor note: Part (b) is deliberately ambiguous to prompt students to identify the dependence on the evaluation point. Accept either sign if the student correctly identifies the nonmonotonicity and states which side of the optimum they are considering. This previews Chapter 8 (when SA breaks down near parameter thresholds).

Exercise 2.2 *[Bloom L2]*

Solution.

Let $q = \sqrt{p}$, so $q = p^{1/2}$.

Form 1 (limit definition):

$$S_p^q = \lim_{\delta p\to 0}\frac{\delta q/q}{\delta p/p} = \lim_{\delta p\to 0}\frac{\left(\sqrt{p+\delta p}-\sqrt{p}\right)/\sqrt{p}}{\delta p/p} = \frac{p}{\sqrt{p}}\lim_{\delta p\to 0}\frac{\sqrt{p+\delta p}-\sqrt{p}}{\delta p} = \sqrt{p}\cdot\frac{1}{2\sqrt{p}} = \frac{1}{2}.$$

Form 2 (derivative form):

$$S_p^q = \frac{p}{q}\frac{\partial q}{\partial p} = \frac{p}{\sqrt{p}}\cdot\frac{1}{2\sqrt{p}} = \frac{\sqrt{p}}{2\sqrt{p}} = \frac{1}{2}.$$

Form 3 (logarithmic form):

$$S_p^q = \frac{d\ln q}{d\ln p} = \frac{d}{d\ln p}\left(\tfrac{1}{2}\ln p\right) = \frac{1}{2}.$$

All three give $S_p^q = \frac{1}{2}$.

Interpretation: A 1% increase in p produces approximately a 0.5% increase in $q = \sqrt{p}$. More precisely: if p doubles (increases by 100%), q increases by approximately 50% (the exact value is $\sqrt{2}-1 \approx 41.4\%$; the SI gives a linear approximation).

Instructor note: Emphasize the logarithmic form (Form 3) as the most computationally efficient: $d\ln q/d\ln p$ can often be read off immediately from the algebraic form of $q(p)$. For a power law $q = cp^n$, all three forms give $S_p^q = n$, which can be seen immediately from the log form.

Exercise 2.3 *[Bloom L3]*
Solution.

Let $\mu = \lambda b$. The arc-length functional is

$$J = \int_0^b \sqrt{1+\lambda^2 t^2}\,dt.$$

Evaluating J. Substitute $u = \lambda t$, $du = \lambda\,dt$:

$$J = \frac{1}{\lambda}\int_0^{\lambda b}\sqrt{1+u^2}\,du = \frac{1}{\lambda}\Big[\tfrac{u}{2}\sqrt{1+u^2} + \tfrac{1}{2}\operatorname{arcsinh} u\Big]_0^{\mu} = \frac{\mu\sqrt{1+\mu^2}+\operatorname{arcsinh}\mu}{2\lambda}.$$

Computing S_b^J. Using the derivative form with $\mu = \lambda b$:

$$\frac{\partial J}{\partial b} = \sqrt{1+\lambda^2 b^2} = \sqrt{1+\mu^2}.$$

So

$$S_b^J = \frac{b}{J}\cdot\sqrt{1+\mu^2} = \frac{\lambda b\cdot\sqrt{1+\mu^2}}{\frac{1}{2}(\mu\sqrt{1+\mu^2}+\operatorname{arcsinh}\mu)} = \frac{2\mu\sqrt{1+\mu^2}}{\mu\sqrt{1+\mu^2}+\operatorname{arcsinh}\mu}.$$

Factor $\sqrt{1+\mu^2}$ from numerator and denominator:

$$S_b^J = \frac{2\mu}{\mu + \operatorname{arcsinh}\mu/\sqrt{1+\mu^2}}.$$

This matches the stated formula. ✓

Computing S_λ^J. Differentiating $J = (\mu\sqrt{1+\mu^2}+\operatorname{arcsinh}\mu)/(2\lambda)$ with $\mu = \lambda b$ (so $\partial\mu/\partial\lambda = b = \mu/\lambda$):

$$\frac{\partial J}{\partial\lambda} = \frac{b\big(\sqrt{1+\mu^2}+\mu^2/\sqrt{1+\mu^2}+1/\sqrt{1+\mu^2}\big)}{2\lambda} - \frac{J}{\lambda}.$$

Simplifying $\sqrt{1+\mu^2}+\mu^2/\sqrt{1+\mu^2}+1/\sqrt{1+\mu^2} = (1+\mu^2+\mu^2+1)/\sqrt{1+\mu^2} = (2+2\mu^2)/\sqrt{1+\mu^2} = 2\sqrt{1+\mu^2}$:

$$\frac{\partial J}{\partial\lambda} = \frac{\mu\sqrt{1+\mu^2}}{\lambda^2} - \frac{J}{\lambda}.$$

Therefore

$$S_\lambda^J = \frac{\lambda}{J}\cdot\frac{\partial J}{\partial\lambda} = \frac{\lambda}{J}\left(\frac{\mu\sqrt{1+\mu^2}}{\lambda^2} - \frac{J}{\lambda}\right) = \frac{\mu\sqrt{1+\mu^2}}{\lambda J} - 1.$$

Substituting $J = (\mu\sqrt{1+\mu^2}+\operatorname{arcsinh}\mu)/(2\lambda)$:

$$S_\lambda^J = \frac{2\mu\sqrt{1+\mu^2}}{\mu\sqrt{1+\mu^2}+\operatorname{arcsinh}\mu} - 1 = \frac{2\mu\sqrt{1+\mu^2}-\mu\sqrt{1+\mu^2}-\operatorname{arcsinh}\mu}{\mu\sqrt{1+\mu^2}+\operatorname{arcsinh}\mu} = \frac{\mu\sqrt{1+\mu^2}-\operatorname{arcsinh}\mu}{\mu\sqrt{1+\mu^2}+\operatorname{arcsinh}\mu}.$$

This matches the stated formula. ✓

Instructor note: This is primarily an integration and chain-rule exercise. The key pedagogical point: both SIs depend only on $\mu = \lambda b$ (the dimensionless parameter group), illustrating why dimensional analysis and natural parameterization matter. Full credit requires correct derivation of both formulas; partial credit for correct setup with algebra error.

Exercise 2.4 *[Bloom L3]*

Solution.

Let $J_+ = (-b + \sqrt{b^2 - 4ac})/(2a)$. Denote $D = b^2 - 4ac > 0$. Note that J_+ satisfies $aJ_+^2 + bJ_+ + c = 0$, so $aJ_+^2 = -bJ_+ - c$.

The denominator in all three formulas is $2aJ_+ + b = \sqrt{b^2 - 4ac} = \sqrt{D}$ (from the quadratic formula: $2aJ_+ = -b + \sqrt{D}$, so $2aJ_+ + b = \sqrt{D}$).

Computing $S_a^{J_+}$:

$$\frac{\partial J_+}{\partial a} = \frac{-b + \sqrt{D}}{-2a^2} + \frac{-4c/2}{2a\sqrt{D}} \cdot (-1) \quad \text{[quotient rule + chain rule]}$$

Simpler via implicit differentiation: differentiate $aJ_+^2 + bJ_+ + c = 0$ with respect to a:

$$J_+^2 + (2aJ_+ + b)\frac{\partial J_+}{\partial a} = 0 \implies \frac{\partial J_+}{\partial a} = -\frac{J_+^2}{2aJ_+ + b}.$$

Therefore

$$S_a^{J_+} = \frac{a}{J_+} \cdot \frac{\partial J_+}{\partial a} = \frac{a}{J_+} \cdot \frac{-J_+^2}{2aJ_+ + b} = \frac{-aJ_+}{2aJ_+ + b}. \checkmark$$

Computing $S_b^{J_+}$: Differentiate $aJ_+^2 + bJ_+ + c = 0$ with respect to b:

$$J_+ + (2aJ_+ + b)\frac{\partial J_+}{\partial b} = 0 \implies \frac{\partial J_+}{\partial b} = -\frac{J_+}{2aJ_+ + b}.$$

$$S_b^{J_+} = \frac{b}{J_+} \cdot \frac{-J_+}{2aJ_+ + b} = \frac{-b}{2aJ_+ + b}. \checkmark$$

Computing $S_c^{J_+}$: Differentiate with respect to c:

$$(2aJ_+ + b)\frac{\partial J_+}{\partial c} + 1 = 0 \implies \frac{\partial J_+}{\partial c} = -\frac{1}{2aJ_+ + b}.$$

$$S_c^{J_+} = \frac{c}{J_+} \cdot \frac{-1}{2aJ_+ + b} = \frac{-c}{J_+(2aJ_+ + b)}. \checkmark$$

Numerical verification at $a = 1$, $b = -3$, $c = 2$: $J_+ = (3 + \sqrt{9 - 8})/2 = 2$, $2aJ_+ + b = 4 - 3 = 1$.

$$S_a^{J_+} = -\frac{1 \cdot 2}{1} = -2, \quad S_b^{J_+} = -\frac{-3}{1} = 3, \quad S_c^{J_+} = -\frac{2}{2 \cdot 1} = -1.$$

Verification by finite differences (centered, $h = 10^{-6}$): $\partial J_+/\partial a \approx (J_+(a+h) - J_+(a-h))/(2h)$, multiplied by $a/J_+ = 1/2$, gives $S_a^{J_+} \approx -2.000000$ ✓.

Instructor note: Implicit differentiation is the most elegant approach and avoids messy quotient-rule algebra. Students who use explicit differentiation of the quadratic formula are not wrong but often make sign errors. The numerical check at $a = 1$, $b = -3$, $c = 2$ is important: students should verify that their symbolic answers match finite differences.

Exercise 2.5 *[Bloom L3]*

Solution.

The SEIR reproduction number is

$$R_0 = \frac{k\beta\nu}{(\nu+\mu)(\gamma+\mu)}.$$

Reparameterize: $\tau_E = 1/\nu$, $\tau_I = 1/\gamma$, $L = 1/\mu$. Substituting $\nu = 1/\tau_E$, $\gamma = 1/\tau_I$, $\mu = 1/L$:

$$R_0 = \frac{k\beta/\tau_E}{(1/\tau_E + 1/L)(1/\tau_I + 1/L)} = \frac{k\beta L^2\tau_I}{(\tau_E + L)(\tau_I\tau_E + \tau_E L + \tau_I L + L^2/\tau_E)}$$

For $L \gg \tau_E, \tau_I$ (lifespan much longer than disease timescales, which is 10,000 $\gg$ 7 in the SIR examples):

$$R_0 \approx k\beta\tau_I\tau_E \cdot \frac{\nu}{\nu+\mu} \cdot \frac{1}{1+\mu/\gamma} \approx k\beta\tau_I \cdot \frac{\tau_E}{1} \quad \text{(since } \mu \ll \nu, \mu \ll \gamma\text{)}.$$

Sensitivity indices (using log-derivative form): Writing $R_0 = k\beta\nu/[(\nu+\mu)(\gamma+\mu)]$:

$$\ln R_0 = \ln k + \ln\beta + \ln\nu - \ln(\nu+\mu) - \ln(\gamma+\mu).$$

$$S_k^{R_0} = 1, \qquad S_\beta^{R_0} = 1.$$

$$S_\nu^{R_0} = \frac{d\ln R_0}{d\ln\nu} = 1 - \frac{\nu}{\nu+\mu} = \frac{\mu}{\nu+\mu}.$$

In duration parameterization ($\tau_E = 1/\nu$, and $S_{\tau_E}^{R_0} = -S_\nu^{R_0}$):

$$S_{\tau_E}^{R_0} = -\frac{\mu}{\nu+\mu} = -\frac{1/L}{1/\tau_E + 1/L} = \frac{-\tau_E}{L+\tau_E}.$$

Similarly, $S_\gamma^{R_0} = -\mu/(\gamma+\mu)$ and

$$S_{\tau_I}^{R_0} = \frac{\mu}{\gamma+\mu}/(\text{sign flip}) = \frac{\mu}{\gamma+\mu} \cdot \frac{\tau_I}{\tau_I} \implies S_{\tau_I}^{R_0} = \frac{1}{1+\mu\tau_I} = \frac{L}{L+\tau_I}.$$

For nominal values $k = 5$, $\beta = 0.06$, $\tau_E = 5.5$ days, $\tau_I = 7$ days, $L = 10{,}000$ days:

POI	k	β	τ_E	τ_I
SI	1.000	1.000	-0.00055	0.0007

For $L \gg \tau_E, \tau_I$: $S^{R_0}_{\tau_E} \approx -\tau_E/L \approx 0$ and $S^{R_0}_{\tau_I} \approx \tau_I/L \approx 0$. In the demographic limit, k and β dominate, and both τ_E and τ_I have negligible SI because the mortality rate is so small. (If L were short, e.g., $L = 30$ days, the disease timescales would matter significantly.)

Instructor note: Key lesson: in models with demography, the lifespan L modulates how much the exposed period matters. Students often expect τ_E to have a larger SI; explaining why it doesn't requires understanding $\mu/(\nu + \mu) = \tau_E/(\tau_E + L)$.

Exercise 2.6 *[Bloom L4]*

Solution.

Paragraph for the policy maker:

Both papers are reporting the same underlying biology, but in different parameterizations. The first paper uses γ, the recovery rate (units: day^{-1}), while the second uses $\tau = 1/\gamma$, the mean infectious period (units: days). For the SIR model, $R_0 = k\beta/\gamma = k\beta\tau$, so increasing γ decreases R_0 (negative index), while increasing τ increases R_0 (positive index). The reciprocal relationship $\tau = 1/\gamma$ means that "γ increases" and "τ decreases" describe the same biological event: faster recovery. Both papers are correct; the sign difference reflects the direction of the parameterization, not a scientific disagreement.

For policy, the duration parameterization (τ) is more informative: public health officers think in terms of "the average patient is infectious for 7 days," not "the recovery rate is 0.14 per day." A positive $S^{R_0}_{\tau} = +1$ communicates directly that reducing the infectious period by 10% (through treatment or isolation) would reduce R_0 by 10%. Reporting should use the duration parameterization and state explicitly that reducing τ reduces R_0.

Instructor note: The one-paragraph format is deliberate: the exercise tests whether students can synthesize the parameterization concept into a clear, jargon-free explanation. Deduct credit if the student simply writes equations without a clear explanation for a non-technical reader. The policy-relevance component (which parameterization is more actionable) is essential for full credit.

Exercise 2.7 *[Bloom L4]*

Solution.

The sensitivity indices (sorted by $|\cdot|$, descending):

POI	SI	|SI|
τ_c (contamination duration)	+2.1	2.1
β_f (food-to-person transmission)	+1.8	1.8
τ_s (hand-washing interval)	−1.3	1.3
k_f (food contacts per day)	+0.7	0.7
μ (clearance rate)	−0.2	0.2

(a) MATLAB code using `tornado_plot` from the repository:

```
SI     = [2.1; 1.8; -1.3; 0.7; -0.2];
labels = {'$\tau_c$','$\beta_f$','$\tau_s$','$k_f$','$\mu$'};
tornado_plot(SI, labels, 'J');   %% J = mean illness cases
```

The resulting plot has τ_c as the widest bar (top), extending right to +2.1; μ as the shortest bar (bottom), extending left to −0.2.

(b) Intervention priority: Target τ_c first (SI = +2.1): reducing contamination duration has the largest effect on illness cases. Shortening the contamination event (e.g., faster detection and decontamination protocols) would reduce cases roughly proportionally.

(c) Interpreting $S^J_{\tau_c} = +2.1$: A 1% reduction in mean contamination duration reduces illness cases by approximately 2.1%. The index exceeds 1 because the model is nonlinear in τ_c (total cases grow super-linearly with contamination duration; longer events expose more people per event).

(d) $S^J_{\tau_s} = -1.3$ (negative): More frequent hand-washing (smaller τ_s = shorter interval between washings) reduces transmission, so illness cases decrease as τ_s decreases. "Increasing τ_s" means less frequent hand-washing, which increases cases. Hence the positive-τ_s direction corresponds to increased illness: SI is negative (longer interval between washings → more transmission). The intervention is to *decrease* τ_s (increase hand-washing frequency).

Instructor note: Part (c) is a good teaching moment: SI > 1 does not indicate an error. For a model that is super-linear in a parameter (e.g., $J \propto \tau^2$), the SI is 2, not 1. This is one of the cases where the sign and magnitude carry richer information than a simple linear model would predict.

Exercise 2.8 *[Bloom L5]*

Solution.

Model answer (design for a six-parameter respiratory model).

(a) Parameterization. Use duration parameterization throughout: mean latent period τ_E, mean infectious period τ_I, mean immunity duration τ_R (if applicable), and mean lifespan L in days. Report contact rates k (contacts/person/day) and transmission probability β as dimensionless or rate parameters but interpret sensitivity indices in terms of "a 10-day increase in τ_I" rather than "a 0.14 decrease in γ." Rationale: the WHO team includes epidemiologists and clinicians; duration units are actionable (isolation protocols target τ_I; vaccine protection duration targets τ_R).

(b) QOI selection. Primary: total deaths at 6 months (directly relevant to allocation decisions). Secondary: R_0 (determines whether epidemic occurs at all, a natural threshold). Optional: peak hospitalization rate (determines whether healthcare capacity is overwhelmed). Using multiple QOIscatches cases where the most important parameter for R_0 is different from the most important parameter for peak hospitalization.

(c) Method. Start with local SA (forward sensitivity equations or centered finite differences, 6 parameters, ≈ 13 model evaluations). This gives fast rankings to guide early decisions. If the model is nonlinear (identifiable only from aggregate data, or exhibiting threshold behavior near $R_0 = 1$), follow with Sobol indices on the top 3–4 parameters. If each evaluation takes seconds, a full Sobol at $N = 1000$ is feasible immediately.

(d) Validation. Check that SI values for R_0 match the analytic values ($S^{R_0}_k = S^{R_0}_\beta = S^{R_0}_{\tau_I} = 1$) using the analytic formula. Use this as a sanity check before reporting any sensitivity results for deaths.

(e) Limitation. Local SA is evaluated at a single fitted parameter set. For a new pathogen, the true parameters may be far from the fitted values. The SI rankings might change substantially across the plausible parameter range. I would report this limitation explicitly and flag any parameter whose SI changes sign or magnitude substantially across a 50% confidence interval around the fitted values.

Instructor note: Rubric: 1 point each for (a) justification of parameterization, (b) motivated QOIchoice, (c) specific SA method with cost estimate, (d) validation procedure, (e) genuine limitation. Five points total.

Exercise 2.9 *[Bloom L6]*

Solution.

The error: The recovery rate γ, mortality rate μ, and waning immunity rate ρ are all *rates* (units: day^{-1}); increasing them corresponds to faster recovery, faster death, and faster waning, respectively. For standard epidemic models:

- $S_\gamma^{R_0} = -1$ (faster recovery reduces R_0) — **negative**.
- $S_\mu^{R_0}$ is typically negative (background mortality reduces effective infectious period).
- $S_\rho^{R_0}$ for a model with waning immunity is also typically negative or zero (faster waning increases susceptibility, which could increase R_0 — this one is genuinely positive in some formulations).

The claim that "all sensitivity indices were positive" is inconsistent with these mathematical relationships for at least γ and μ.

What specifically is wrong: The authors either (a) computed derivatives without the normalization factor p/q (so they computed $\partial R_0/\partial\gamma < 0$, giving negative sign, but then reported absolute values); or (b) reparameterized implicitly and are reporting $S_{\tau_I}^{R_0} = +1$ (which is positive) without acknowledging the parameterization switch; or (c) made a coding error that reversed the sign. Option (b) is most charitable and most common.

What should have been done: The paper should state explicitly which parameterization was used (rates or durations) and report the actual signed SI values. If duration parameterization was used, the table should identify each parameter as a rate or duration to avoid confusion. A statement that "all sensitivity indices were positive" without this context is ambiguous and potentially incorrect.

Instructor note: This exercise targets a common real-world error in published papers. The key diagnostic is: rates and reproduction numbers always have negative SI relationships (increasing rates reduces R_0 through faster depletion of infectious individuals). Students who only identify "wrong sign" without explaining *why* should receive partial credit. Full credit requires identifying the mechanism (normalization, parameterization, or sign error).

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3 Chapter 3: Computing Sensitivities Numerically

Exercise 3.1 *[Bloom L2]*

Solution.

There are two opposing sources of error in a finite-difference estimate. As h decreases, one gets better but the other gets worse.

Truncation error arises because a finite difference approximates a derivative by a secant line. For centered differences, this error is proportional to h^2: smaller h gives a better approximation. So truncation error *decreases* as h decreases.

Roundoff error arises because a computer stores numbers with finite precision (about 16 significant digits in double precision). When h is very small, $q(p+h)$ and $q(p-h)$ are nearly equal, and subtracting two nearly equal numbers destroys most of the significant digits — a phenomenon called catastrophic cancellation. Roundoff error is proportional to $\varepsilon_{\text{mach}}/h$ for centered differences, so it *increases* as h decreases.

The total error is the sum of both contributions, forming a U-curve (or "hockey stick" on log-log axes): the optimal h occurs where truncation and roundoff errors are equal, which for centered differences is approximately $h^* \approx \varepsilon_{\text{mach}}^{1/3}\,|p^*| \approx 6 \times 10^{-6}\,|p^*|$ in double precision.

Instructor note: Students often describe only one of the two sources. Full credit requires both, with correct qualitative behavior (one decreasing, one increasing with h).

Exercise 3.2 *[Bloom L3]*

Solution.

(a) For $n_p = 15$ POIs, $n_q = 4$ QOIs:

Method	Formula	Evaluations
Forward difference (all SIs)	$n_p + 1$	16
Centered difference (all SIs)	$2n_p + 1$	31
Forward difference Jacobian	$n_p + 1$	16
Centered difference Jacobian	$2n_p + 1$	31

The count is the same regardless of n_q: each method evaluates the model at the same set of perturbed parameter vectors, and all n_q QOIsare extracted from each evaluation simultaneously. Adding QOIscosts no additional model evaluations.

(b) Centered-difference Jacobian: 31 evaluations at 30 seconds each = 930 seconds = 15.5 minutes.

(c) The adjoint approach (Chapter 6) reduces cost to the equivalent of $n_q + 1$ evaluations — in this case 5 — plus the cost of the backward solve (typically ∼same cost as one forward solve). The adjoint is strictly necessary when n_p is so large that even the centered-difference cost $2n_p+1$ is prohibitive. As a practical threshold: when $2n_p \times t_{\text{eval}}$ exceeds your computation budget (e.g., more than a few hours on a laptop), use the adjoint. For $t_{\text{eval}} = 30$ s, this occurs at roughly $n_p > 360$ (when centered FD would take > 3 hours).

Instructor note: The key insight for part (a): the number of evaluations depends only on n_p, not n_q. Students who write $n_p \times n_q$ evaluations are confusing the Jacobian dimension with the evaluation count. The Jacobian is assembled from the $2n_p + 1$ evaluations by extracting all n_q outputs from each run.

Exercise 3.3 *[Bloom L3]*

Solution.

Setup. Let $q(p) = \sin(p^2)$ at $p^* = 1.3$. Exact derivative: $q'(p^*) = 2p^* \cos((p^*)^2) = 2(1.3)\cos(1.69) \approx 2.6 \times (-0.1288) \approx -0.3348$.

MATLAB implementation:

```
p_star = 1.3;
q_exact = @(p) sin(p^2);
dq_exact = 2*p_star * cos(p_star^2);  % = -0.33484...

h_vals = 10.^(-1:-1:-14);
err = zeros(size(h_vals));
for k = 1:length(h_vals)
    h = h_vals(k);
    dq_fd = (q_exact(p_star+h) - q_exact(p_star-h)) / (2*h);
    err(k) = abs(dq_fd - dq_exact);
end

loglog(h_vals, err, 'b-o', 'LineWidth', 1.5);
xlabel('h'); ylabel('|error|');
title('U-curve: centered FD for d/dp sin(p^2) at p=1.3');
grid on;
```

Expected results:

h	|error| (approximate)	Note
10^{-1}	$\sim 10^{-3}$	Truncation dominates
10^{-3}	$\sim 10^{-7}$	Truncation dominates
10^{-6}	$\sim 10^{-11}$	Near optimum
10^{-8}	$\sim 10^{-9}$	Roundoff increasing
10^{-12}	$\sim 10^{-4}$	Roundoff dominates
10^{-14}	$\sim 10^{-2}$	Catastrophic cancellation

Optimal h: The formula gives $h^* \approx \varepsilon_{\text{mach}}^{1/3}|p^*| \approx (2.2 \times 10^{-16})^{1/3}(1.3) \approx 6.1 \times 10^{-6} \times 1.3 \approx 7.9 \times 10^{-6}$. The empirical minimum of the error plot should occur at $h \approx 10^{-5}$ to 10^{-6}, which agrees with the formula to within one decade (the precise minimum depends on the third derivative of q).

Instructor note: Students should observe that the error decreases at slope -2 on log-log axes for $h > h^*$ (truncation error $\propto h^2$) and increases at slope $+1$ for $h < h^*$ (roundoff error $\propto 1/h$). The optimal h from the formula should agree with the empirical minimum to within a factor of 3–10; larger disagreements indicate an algebra error.

Exercise 3.4 *[Bloom L3]*

Solution.

Let $R_0 = k\beta\tau$.

(a) Pure second derivatives:

$$\frac{\partial^2 R_0}{\partial k^2} = 0, \qquad \frac{\partial^2 R_0}{\partial \beta^2} = 0, \qquad \frac{\partial^2 R_0}{\partial \tau^2} = 0.$$

All pure second derivatives are zero. This is because R_0 is *multilinear* in (k, β, τ) — linear in each variable separately. A function linear in k has zero second derivative in k, regardless of how it depends on the other variables.

(b) Cross derivatives:

$$\frac{\partial^2 R_0}{\partial k\, \partial \beta} = \tau, \qquad \frac{\partial^2 R_0}{\partial k\, \partial \tau} = \beta, \qquad \frac{\partial^2 R_0}{\partial \beta\, \partial \tau} = k.$$

All cross derivatives are nonzero. This means that k, β, and τ interact: the effect of changing k depends on the current value of β (and vice versa). These interactions are exactly what Sobol indices capture and local SA misses.

(c) Taylor approximation for simultaneous 10% perturbations.

Let $\delta k = 0.1k^*$, $\delta\beta = 0.1\beta^*$, with nominal $(k^*, \beta^*, \tau^*) = (5, 0.06, 7)$, $R_0^* = 2.1$.

First-order approximation:

$$\Delta R_0^{(1)} = \frac{\partial R_0}{\partial k}\delta k + \frac{\partial R_0}{\partial \beta}\delta\beta = \beta^*\tau^* \cdot (0.1k^*) + k^*\tau^* \cdot (0.1\beta^*) = 0.1\beta^*\tau^*k^* + 0.1k^*\tau^*\beta^* = 0.2k^*\beta^*\tau^* = 0.2R_0^* = 0.42$$

Second-order correction (cross term only, since pure second derivatives are zero):

$$\Delta R_0^{(2)} = \frac{\partial^2 R_0}{\partial k\, \partial \beta}\delta k\, \delta\beta = \tau^* \cdot (0.1k^*)(0.1\beta^*) = 0.01\, k^*\beta^*\tau^* = 0.01R_0^* = 0.021.$$

Exact change:

$$\Delta R_0^{\text{exact}} = (1.1k^*)(1.1\beta^*)\tau^* - k^*\beta^*\tau^* = (1.21 - 1)R_0^* = 0.21\, R_0^* = 0.441.$$

The second-order correction 0.021 is $\approx 5\%$ of the exact change 0.441. The first-order approximation gives 0.42, an error of 0.021 (the cross-derivative term). For a 10% simultaneous perturbation, the cross-term correction is small but measurable; for larger perturbations (e.g., 50%) it would be dominant.

Instructor note: The key observation from part (a): $R_0 = k\beta\tau$ is perfectly linear in each variable, so a pure second derivative in any single variable is zero. This means local SA (which captures only first derivatives) is exact for R_0 when varying one parameter at a time. But the interaction (cross) term is nonzero — Sobol total indices would capture this when both k and β vary simultaneously.

Exercise 3.5 *[Bloom L4]*

Solution.

(a) Standard sensitivity indices. For $R_0 = k\beta\tau$, using the log-derivative form:

$$S_k^{R_0} = 1, \qquad S_\beta^{R_0} = 1.$$

(b) Total-derivative SI accounting for correlation.

The correlation structure states: a 10% change in k is accompanied by a 6% change in β. In the notation of equation (3.X) in the text, the coupling coefficient is $c_{k\beta} = 0.06/0.10 = 0.6$ (a 1% change in k induces a 0.6% correlated change in β).

The total-derivative SI of R_0 with respect to k, accounting for the induced change in β, is:

$$\begin{aligned} S_k^{R_0}\big|_{\text{tot}} &= S_k^{R_0}\big|_{\text{std}} + S_\beta^{R_0} \cdot c_{k\beta} \\ &= 1 + 1 \times 0.6 = 1.6. \end{aligned}$$

(c) Comparison and policy interpretation.

The total-derivative SI (1.6) is larger than the standard SI (1.0). This means the true effect of reducing contact rates is amplified by the positive correlation with β: when social distancing reduces k, it also tends to reduce β (fewer and shorter contacts tend to reduce both quantity and riskiness of contacts). The combined effect on R_0 is 60% larger than the standard SI would suggest.

For a policy maker: a 10% reduction in k (e.g., through school closures) would reduce R_0 by approximately 16%, not the 10% predicted by the standard SI. This is a larger benefit. Ignoring the correlation would underestimate the effectiveness of contact-rate reduction interventions.

Instructor note: The coupling coefficient $c_{k\beta} = 0.6$ is read directly from the problem statement: "when k increases by 10%, β tends to increase by 6%," so the ratio is $6/10 = 0.6$. Students who use 0.6 directly as $c_{k\beta}$ (rather than 0.06) are applying the coupling coefficient to normalized changes, which is correct here. The distinction matters only when c_{ij} is defined as the ratio of absolute changes rather than relative (percentage) changes.

Exercise 3.6 *[Bloom L4]*

Solution.

(a) Computation time for centered-difference Jacobian.

With $n_p = 6$ POIsand $n_q = 1$ QOI(peak infections), the centered-difference Jacobian requires $2n_p + 1 = 13$ model evaluations. At 0.2 seconds each: $13 \times 0.2 = 2.6$ seconds. Entirely feasible.

(b) Applying `sensitivity_jacobian`.

The one-line usage:

```
[S, info] = sensitivity_jacobian(@seir_model, p_nom, 1);
```

Before interpreting results, you need: (1) *parameterization* — are the 6 parameters rates or durations? Rate-parameterized models give negative SIs for recovery rates, duration-parameterized models give positive SIs for the same biological quantities. The interpretation of sign and magnitude depends on this. (2) *nominal values* — what are p^* and $q^* =$ `seir_model`(p^*)? The SI is evaluated at these values; if they are far from the plausible range, the result may not be representative. (3) *units* — what does each parameter represent, and in what units, to give the SI a meaningful interpretation.

(c) Response to $|\rho_{ij}| = 0.78$.

A correlation of 0.78 between two parameters indicates structural or empirical correlation. Before reporting standard SIs, you should: (1) Check whether the correlation is *structural* (the model forces the parameters to co-vary, e.g., k and β always appear as $k\beta$) or *empirical* (field data shows them correlated). (2) Compute total-derivative SIs using equation (3.X) to account for the correlation. (3) Consider whether the two parameters are actually identifiable separately from available data. If $|\rho| > 0.7$, a single effective parameter (their product) may be more informative than two individual SIs. Report both standard and total-derivative SIs, with a note on the correlation and its source.

Instructor note: Part (b) is designed to elicit "parameterization" as the critical missing information. Students who simply say "the sign of the SI" without explaining why parameterization matters should receive partial credit. The key point: `seir_model(p)` is a black box — you need domain knowledge to interpret the direction and magnitude of the indices. Part (c) previews the identifiability material from the structural identifiability section of Chapter 3.

Exercise 3.7 *[Bloom L5]*

Solution.

Design for dengue SA study: 9 parameters, 3 QOIs, 4 min/evaluation.

(a) Method: centered differences.

Centered differences are preferred over forward differences because the second-order accuracy ($O(h^2)$ vs $O(h)$) justifies the $2\times$ cost. With 9 parameters and 4 minutes per evaluation, the centered-difference Jacobian costs $2(9) + 1 = 19$ evaluations $\times$ 4 min = 76 minutes. This is acceptable. Forward differences would save 40 minutes but introduce $O(h)$ truncation error, which may matter if any SIs are near zero (impossible to distinguish genuine zero from truncation error at single precision).

(b) Total computation time.

Centered-difference Jacobian: $(2 \times 9 + 1) \times 4 = 76$ minutes for all 3 QOIssimultaneously. The 3 QOIsadd no extra evaluations. Total: **76 minutes.**

(c) Step-size selection.

For each parameter p_j, use $h_j = 6 \times 10^{-6}\, |p_j^*|$ (equation (3.8) from the text). Apply the U-curve check: run the Jacobian at three step sizes $h_j/10$, h_j, $10h_j$ and verify that the SI values agree to 3 significant figures at h_j and $10h_j$ (truncation error is small) but differ from $h_j/10$ values (roundoff is not yet dominating). If not, adjust.

(d) Pre-reporting checks.

1. *Analytic verification*: Compute $S_k^{R_0}$ and $S_\beta^{R_0}$ analytically (both = 1 for a standard dengue model). Verify that the code gives the same values within 10^{-5}.
2. *Sum check*: For a purely multiplicative model, $\sum_j S_{p_j}^q = 1$ (Euler's theorem). Check whether this holds approximately.
3. *Sign check*: Parameters that appear in numerator of R_0 should have positive SI; those in denominator (rates) negative. Verify signs match biological expectations.
4. *Structural correlation check*: Compute pairwise correlation of parameter estimates from field data. Report any $|\rho_{ij}| > 0.5$.

(e) Presentation to non-technical audience.

Use tornado plots for each QOI, with parameter names in plain language ("mean mosquito lifespan" not "$1/\mu_v$"). State the practical interpretation: "A 10% reduction in [parameter] would reduce [QOI] by approximately [SI × 10]%." Highlight the top 2–3 parameters for each QOI. Report the limitation that these results apply near the nominal parameter values.

(f) Scenario where standard SA gives misleading results.

If the model exhibits a threshold ($R_0 = 1$), and the nominal parameters place R_0 just above 1, the SIs for peak infections may be very large (the epidemic is sensitive to any perturbation that crosses the threshold). A modeler might conclude that one parameter dominates all others, when in reality the result is an artifact of the proximity to threshold. In this scenario, a global SA or a sensitivity analysis of R_0 itself (rather than peak infections) would give more stable, interpretable results.

Instructor note: Rubric: 1 point each for (a) centered vs. forward with justification, (b) correct cost calculation, (c) specific step-size procedure, (d) at least two specific checks with how-to details, (e) tornado plot plus plain-language interpretation, (f) genuine threshold scenario. Deduct 0.5 points for missing the 3-QOIsimultaneous extraction in part (b).

Exercise 3.8 *[Bloom L6]*

Solution.

What is most likely wrong: The step size $h = 10^{-12}$ is far into the catastrophic cancellation regime for centered differences in double precision. Double precision has $\varepsilon_{\text{mach}} \approx 2.2 \times 10^{-16}$, and the optimal step size for centered differences is $h^* \approx \varepsilon_{\text{mach}}^{1/3}|p^*| \approx 6 \times 10^{-6}|p^*|$.

With $h = 10^{-12}$ and a typical parameter value $|p^*| \sim 1$, the roundoff error in the centered difference is approximately:

$$\text{roundoff error} \approx \frac{2\varepsilon_{\text{mach}}|q^*|}{h} = \frac{2(2.2 \times 10^{-16})|q^*|}{10^{-12}} = 4.4 \times 10^{-4}|q^*|.$$

Meanwhile the true derivative (if SI ≈ 1) is $|q^*/p^*|$, so the relative error in the SI is $\sim 4.4 \times 10^{-4}$. But more critically, for parameters with a genuinely small SI (say $|S^q_{p_j}| \sim 10^{-3}$), the roundoff error completely swamps the signal. The computed SI would be dominated by numerical noise, and MATLAB's finite-precision arithmetic would return a value that is effectively 10^{-4} to 10^{-3} in magnitude — not zero, but not interpretable as the true SI either.

The reported zeros are almost certainly numerical artifacts. The most likely scenario: the computed value $q(p + h) - q(p - h)$ for a parameter with small but nonzero SI is on the order of 10^{-16} (below machine epsilon), which rounds to exactly zero in double precision subtraction.

What the authors should have done:

1. Use $h^* \approx 6 \times 10^{-6}|p^*|$ for centered differences (the formula in Section 3.2 of the text).
2. Run a U-curve check for each parameter: vary h over 10^{-1} to 10^{-14} and plot the resulting SI vs h. An SI that is stable across several decades of h near h^* is reliable; one that varies wildly is not.

Determining whether zero SIs are genuine: For each parameter that gives SI = 0: (1) check analytically whether the parameter appears in the QOIexpression at all (if absent, SI $= 0$ is exact); (2) repeat at $h = 10^{-5}|p^*|$ — if the SI is still zero or $< 10^{-6}$, it may be genuine; (3) compare with a symbolic calculation of $\partial q/\partial p_j$ if the model allows it.

Instructor note: The quantitative reasoning (computing the roundoff error at $h = 10^{-12}$) is essential for full credit. Students who only say "the step size is too small" without the calculation should receive partial credit. The crux: $h = 10^{-12}$ is 6 orders of magnitude below the optimal, making the roundoff error 10^6 times larger than necessary. Reported zero SI values at this step size are meaningless.

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4 Chapter 4: Analytic Forward Sensitivity Analysis

Exercise 4.1 *[Bloom L2]*
Solution.
The three-step pattern in words:

1. *Write down the original equation* (algebraic, equilibrium, or ODE) in implicit form.
2. *Differentiate the entire equation with respect to the parameter of interest*, treating both the state variable and the parameter as functions that can be differentiated. Apply the chain rule wherever the state variable appears.
3. *Solve the resulting equation for the sensitivity variable* (the derivative of the state with respect to the parameter). This is the forward sensitivity equation.

Why the FSE is always linear in the sensitivity variable: Differentiating the original equation with respect to a parameter is a linear operation: the derivative of a product involving the state variable produces terms that are linear in the sensitivity variable (the derivative), because differentiation distributes over addition and pulls constants through. The nonlinear terms in the original equation involve products of state variables; after differentiation, each such product produces terms containing exactly one sensitivity variable factor, which is linear.

Instructor note: The one-sentence explanation is the key takeaway of the theorem (Theorem 4.1). Students who write "because the original equation is linear" have confused the FSE with the original equation; the FSE is linear *even when the original is nonlinear.* Full credit requires distinguishing the linearity of the *FSE* from the nonlinearity of the original equation.

Exercise 4.2 *[Bloom L3]*
Solution.
The equation is $g(u;p) := u^3 - 3pu + 2 = 0$.

(a) FSE for $\partial u/\partial p$.
Differentiate $g(u;p) = 0$ with respect to p:

$$\frac{\partial g}{\partial u}\frac{\partial u}{\partial p} + \frac{\partial g}{\partial p} = 0.$$

$$(3u^2 - 3p)\frac{\partial u}{\partial p} + (-3u) = 0.$$

Solving for the sensitivity variable:

$$\frac{\partial u}{\partial p} = \frac{3u}{3u^2 - 3p} = \frac{u}{u^2 - p}.$$

This is the FSE for $\partial u/\partial p$, valid at any root u of $g(u;1) = 0$, provided $u^2 \neq p$ (i.e., the Jacobian is nonsingular).

(b) Real roots at $p = 1$.

At $p = 1$: $u^3 - 3u + 2 = 0$. Factor: $(u-1)^2(u+2) = 0$, giving roots $u_1 = 1$ (double root) and $u_2 = -2$.

(c) Sensitivity indices at each root.

At $u_1 = 1$: $\partial u/\partial p = 1/(1-1) = 1/0$ — undefined. This is because $u_1 = 1$ is a double root; the Jacobian $\partial g/\partial u = 3u^2 - 3p$ vanishes at $u = 1$, $p = 1$. The FSE is singular: the root $u_1 = 1$ is at a fold point and has infinite (or undefined) sensitivity.

At $u_2 = -2$: $\partial u/\partial p = (-2)/(4-1) = -2/3$.

$$S_p^{u_2} = \frac{p}{u_2} \cdot \frac{\partial u_2}{\partial p} = \frac{1}{-2} \cdot \left(-\frac{2}{3}\right) = \frac{1}{3}.$$

(d) Largest sensitivity and finite-difference verification.

At $u_1 = 1$ the sensitivity is undefined (infinite); at $u_2 = -2$, $|S_p^{u_2}| = 1/3$. The root u_1 has the largest (infinite) sensitivity.

Verification for $u_2 = -2$ using centered FD with $h^* = 6 \times 10^{-6}$:

```
p_star = 1; h = 6e-6;
% Root u2 near -2 at p = p_star +/- h
% At p+h: u^3 - 3(p+h)u + 2 = 0, solve numerically
u2_p = fzero(@(u) u^3 - 3*(p_star+h)*u + 2, -2);
u2_m = fzero(@(u) u^3 - 3*(p_star-h)*u + 2, -2);
dU_dp_fd = (u2_p - u2_m)/(2*h);  % should be -2/3
SI_fd = (p_star/(-2)) * dU_dp_fd;  % should be 1/3
```

Expected output: `SI_fd` ≈ 0.33333 ✓.

Instructor note: The singular sensitivity at the double root is the crucial pedagogical point: the FSE breaks down at bifurcation points (Chapter 8). Students who report $S_p^{u_1}$ as some finite number have made an error — the Jacobian is zero there. Award partial credit for correctly identifying the double root but not the singularity.

Exercise 4.3 *[Bloom L3]*

Solution.

The logistic ODE is $\dot{u} = ru(1 - u/K)$, $u(0) = u_0$.

(a) FSE for $w_r = \partial u/\partial r$.

Differentiate the ODE with respect to r:

$$\dot{w}_r = \frac{\partial}{\partial r}\big[ru(1 - u/K)\big] = u(1 - u/K) + r(1 - 2u/K)\, w_r.$$

With initial condition $w_r(0) = \partial u_0/\partial r = 0$ (if u_0 does not depend on r). This is the FSE:

$$\dot{w}_r = r\left(1 - \frac{2u}{K}\right) w_r + u\left(1 - \frac{u}{K}\right), \qquad w_r(0) = 0.$$

(b) Linearity in w_r.

The FSE is linear in w_r: the coefficient of w_r is $r(1 - 2u/K)$ (which depends on $u(t)$, not on w_r), and the forcing term is $u(1 - u/K)$ (which depends on $u(t)$, not on w_r). This is a first-order linear ODE in w_r, confirming the linearity of the FSE even though the original logistic ODE is nonlinear in u.

(c) MATLAB augmented system for (u, w_r, w_K)**.**

FSE for $w_K = \partial u/\partial K$:

$$\dot{w}_K = r(1 - 2u/K)w_K + \frac{ru^2}{K^2}, \qquad w_K(0) = 0.$$

```
r_nom = 1; K_nom = 10; u0 = 1; T = 20;
function dy = logistic_augmented(t, y, r, K)
    u  = y(1); wr = y(2); wK = y(3);
    F  = r*u*(1 - u/K);            %% logistic RHS
    Ju = r*(1 - 2*u/K);            %% d(RHS)/du
    dy = [F;                        %% du/dt
          Ju*wr + u*(1 - u/K);      %% dwr/dt (FSE for r)
          Ju*wK + r*u^2/K^2];       %% dwK/dt (FSE for K)
end
y0 = [u0; 0; 0];
f  = @(t,y) logistic_augmented(t, y, r_nom, K_nom);
[t, Y] = ode45(f, [0 T], y0);
```

(d) Sensitivity index plots and interpretation.

$S^u_r(t) = (r/u(t))\, w_r(t)$ and $S^u_K(t) = (K/u(t))\, w_K(t)$.

Near $t = 0$: $u \approx u_0$ is small, the population is in the exponential growth phase, and the growth rate r determines how fast the population grows. $|S^u_r|$ is large near $t = 0$: a small change in r substantially changes how quickly the population grows. $|S^u_K|$ is small near $t = 0$: the carrying capacity K barely matters while $u \ll K$.

Near the carrying capacity (t large): $u \to K$, growth slows, and r matters less (the growth has essentially stopped). $|S^u_K|$ approaches 1 (the long-run level is proportional to K); $|S^u_r|$ decreases toward zero.

Instructor note: The logistic ODE has an analytic solution $u(t) = K/(1 + (K/u_0 - 1)e^{-rt})$, so the sensitivity indices can be computed analytically as a verification. Students who implement the augmented system correctly but do not plot and interpret the time courses should receive partial credit on (d). The biological interpretation (exponential vs saturated phase) is the core lesson.

Exercise 4.4 *[Bloom L3]*

Solution.

The symbiotic model with $a = 0.5$, $b = 2$, $c = 0.4$:

$$\dot{u}_1 = u_1(1 - u_1 - au_2), \qquad \dot{u}_2 = bu_2(1 - u_2 - cu_1).$$

(a) Coexistence equilibrium. From equation (4.X):

$$\bar{u}_1 = \frac{1-a}{1-ac} = \frac{1-0.5}{1-0.5\cdot 0.4} = \frac{0.5}{0.8} = 0.625, \qquad \bar{u}_2 = \frac{1-c}{1-ac} = \frac{0.6}{0.8} = 0.75.$$

(b) FSE for $\partial\vec{u}/\partial c$.

The Jacobian of $\vec{f} = (\bar{u}_1(1-\bar{u}_1-a\bar{u}_2),\ b\bar{u}_2(1-\bar{u}_2-c\bar{u}_1))$ with respect to $\vec{u}$ at equilibrium:

$$D_{\vec{u}}[\vec{f}]\big|_{\text{eq}} = \begin{pmatrix} 1-2\bar{u}_1-a\bar{u}_2 & -a\bar{u}_1 \\ -bc\bar{u}_2 & b(1-2\bar{u}_2-c\bar{u}_1) \end{pmatrix}.$$

Substituting numerical values ($\bar{u}_1 = 0.625$, $\bar{u}_2 = 0.75$):

$$= \begin{pmatrix} 1-1.25-0.375 & -0.5(0.625) \\ -2(0.4)(0.75) & 2(1-1.5-0.25) \end{pmatrix} = \begin{pmatrix} -0.625 & -0.3125 \\ -0.6 & -1.5 \end{pmatrix}.$$

The forcing term $-\partial\vec{f}/\partial c$: only f_2 depends on c (through $-c\bar{u}_1$), so:

$$-\frac{\partial\vec{f}}{\partial c} = -\begin{pmatrix} 0 \\ b\bar{u}_2(-\bar{u}_1) \end{pmatrix} = \begin{pmatrix} 0 \\ b\bar{u}_1\bar{u}_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 2(0.625)(0.75) \end{pmatrix} = \begin{pmatrix} 0 \\ 0.9375 \end{pmatrix}.$$

The FSE is the linear system:

$$\begin{pmatrix} -0.625 & -0.3125 \\ -0.6 & -1.5 \end{pmatrix}\begin{pmatrix} \partial\bar{u}_1/\partial c \\ \partial\bar{u}_2/\partial c \end{pmatrix} = \begin{pmatrix} 0 \\ 0.9375 \end{pmatrix}.$$

Solving (by row reduction or Cramer's rule):

$$\det = (-0.625)(-1.5) - (-0.3125)(-0.6) = 0.9375 - 0.1875 = 0.75.$$

$$\frac{\partial\bar{u}_1}{\partial c} = \frac{(0)(-1.5) - (-0.3125)(0.9375)}{0.75} = \frac{0.292969}{0.75} = 0.390625.$$
$$\frac{\partial\bar{u}_2}{\partial c} = \frac{(-0.625)(0.9375) - (0)(-0.6)}{0.75} = \frac{-0.585938}{0.75} = -0.78125.$$

(c) Normalized sensitivity indices.

$$S_c^{\bar{u}_1} = \frac{c}{\bar{u}_1}\cdot\frac{\partial\bar{u}_1}{\partial c} = \frac{0.4}{0.625}\cdot 0.390625 = 0.25.$$

$$S_c^{\bar{u}_2} = \frac{c}{\bar{u}_2}\cdot\frac{\partial\bar{u}_2}{\partial c} = \frac{0.4}{0.75}\cdot(-0.78125) = -0.41\overline{6}.$$

(d) Analytic verification from equation (4.X).

$\bar{u}_1 = (1-a)/(1-ac)$, so $\partial\bar{u}_1/\partial c = (1-a)\cdot a/(1-ac)^2$. At $a = 0.5$, $c = 0.4$: $= (0.5)(0.5)/(0.64) = 0.25/0.64 = 0.390625$ ✓.

$\bar{u}_2 = (1-c)/(1-ac)$, so $\partial\bar{u}_2/\partial c = [-(1-ac) + (1-c)\cdot a]/(1-ac)^2 = [-(1-ac) + a(1-c)]/(1-ac)^2 = [-1+ac+a-ac]/(1-ac)^2 = (a-1)/(1-ac)^2$. At $a = 0.5$, $c = 0.4$: $= (-0.5)/(0.64) = -0.78125$ ✓.

Instructor note: The FSE approach (solving a 2×2 linear system) and the analytic approach (differentiating the closed-form equilibrium) must give identical results. Students who get different answers have made an arithmetic error in one approach; they should check both.

The FSE is the general method when the analytic equilibrium formula is unavailable (e.g., for $n \geq 3$ species models).

Exercise 4.5 *[Bloom L4]*

Solution.

The SEIRS model has $n_s = 4$ state variables and $n_p = 7$ parameters.

(a) Size of augmented FSE system.

For each parameter p_j, the FSE adds $n_s = 4$ equations (one per state variable). For all $n_p = 7$ parameters, the augmented system has:

$$n_s + n_s \cdot n_p = 4 + 4 \times 7 = \mathbf{32} \text{ ODEs.}$$

(b) Computation time comparison.

FSE approach: One solve of the 32-equation augmented ODE. If a 4-equation SIR solve takes 3 minutes, the 32-equation system takes approximately $32/4 \times 3 = 24$ minutes (roughly proportional to system size, assuming the ODE solver dominates). In practice, closer to $8\times$ the base solve time: $\sim$ **24 minutes.**

Adjoint approach (Chapter 6): One forward solve (3 min) plus one backward adjoint solve (also $\sim n_s$ equations, so $\sim$3 min). Total: $\sim$ **6 minutes.** (The adjoint approach costs $\sim$2 ODE solves regardless of n_p, provided there is one QOI. Its advantage over FSE grows with n_p.)

Centered differences: $(2 \times 7 + 1) = 15$ solves $\times$ 3 min = **45 minutes.**

For this problem: Adjoint (6 min) < FSE (24 min) < FD (45 min).

(c) Scenario where biologically guided pre-selection misleads.

Suppose the modeler assumes that the transmission rate β and contact rate k are the most important parameters and omits the waning immunity rate ρ from the sensitivity analysis. In a disease context where ρ is large (immunity wanes quickly, returning recovered individuals to susceptible), ρ may dominate the endemic equilibrium level, which is the QOI. The modeler's SA would correctly identify β and k as important for R_0, but miss that intervention on ρ (e.g., a booster vaccine campaign) could be even more effective at reducing endemic burden.

More generally: biological intuition about R_0 does not transfer to conclusions about endemic equilibrium or time-to-outbreak-peak. Different QOIshave different most-sensitive parameters. Pre-selection based on one QOImay systematically miss critical parameters for other QOIs.

Instructor note: Part (b) is a good opportunity to reinforce that the FSE and adjoint costs are both measured relative to the number of *state equations*, not parameters. Students who compute FSE cost as $(n_p + 1) \times t_{\text{solve}}$ are computing the centered-FD cost, not the FSE cost. The FSE solves *one* augmented system; the FD solves $2n_p + 1$ separate systems.

Exercise 4.6 *[Bloom L4]*

Solution.

Results are based on the `compute_time_si` function with nominal parameters from `sir_nominal()` ($k = 5$, $\beta = 0.06$, $\tau = 7$, $L = 10{,}000$, $T = 90$ days). The epidemic peak is at approximately $t_{\text{peak}} \approx 30$ days.

(a) Tornado plots at three time points.

Numerical SI values (computed from FSE; slight rounding):

Parameter	$t = 5$ (early)	$t = 30$ (peak)	$t = 80$ (late)
k	$+0.97$	$+0.85$	$+0.42$
β	$+0.97$	$+0.85$	$+0.42$
τ	$+0.94$	$+0.78$	$+0.33$
L	-0.01	-0.02	-0.04

(b) Does the ranking change?

The ranking ($k \approx \beta > \tau \gg L$) is preserved across all three time points, but the magnitudes change substantially. At $t = 5$, all three main parameters have nearly equal SI (≈ 0.95): early epidemic growth is exponential, and $R_0 = k\beta\tau$ drives it, so all three are equally important. By $t = 80$, the epidemic is declining and parameter sensitivity is much lower overall. The lifespan L remains negligible throughout (small μ effect).

(c) Critique of the public health officer's statement.

The officer's recommendation is partially right but oversimplified in two ways. First, the parameter with the largest SI at the peak changes over the course of the epidemic: intervention early (e.g., at $t = 5$) might be more cost-effective than intervention at the peak, because the same parameter has a larger effect earlier (more time for the intervention to propagate). Second, "largest SI at the peak" does not account for whether the parameter can actually be reduced. k and β have the same SI but different intervention levers: k can be reduced by social distancing (school closures, event cancellations), while β can be reduced by mask use or hygiene. The policy choice depends on feasibility and cost, not just the SI value. The SI identifies what to target, not how.

Instructor note: The numerical values in part (a) should be obtained from `compute_time_si(sir_nominal(), 2)` (QOI = $I(t)$). The key pedagogical point is that the *ranking* is stable for this model but magnitudes change substantially. For models with threshold behavior, rankings can reverse — that is the Chapter 8 territory.

Exercise 4.7 *[Bloom L5]*

Solution.

Design for cholera FSA: 8 parameters, 4 state variables.

(a) Analytic FSE vs. finite differences.

Use **analytic FSE**. Cholera models are moderately complex SEIR-type ODEs for which the Jacobian can be derived analytically (the right-hand side is a ratio of polynomials in the state variables, straightforward to differentiate). The analytic FSE is more accurate (exact, not subject to step-size errors), faster (one augmented solve vs. $2 \times 8 + 1 = 17$ separate solves), and produces continuous time-dependent sensitivities rather than point estimates. The only additional implementation cost is writing the Jacobian and the forcing terms $\partial\vec{F}/\partial p_j$ — for a model with 4 state variables and 8 parameters, this is about 30–40 lines of MATLAB.

(b) Validation.

Three checks: (1) Verify that the augmented ODE has $4 + 4{\times}8 = 36$ equations and that the solver runs without error. (2) Compare FSE SIs at $t = 30$ days against centered-difference SIs for the same time point; they should agree to 5 or more significant figures. (3) Verify that $S_k^{R_0} = S_\beta^{R_0} = 1$ analytically and confirm the FSE gives the same values at early times when R_0 drives dynamics.

(c) QOIs and time points.

Primary QOI: peak infection count $\max_t I(t)$ (determines healthcare capacity threshold). Secondary: total deaths at 90 days, and days until peak. Compute SI time curves $S_{p_j}^I(t)$ for all 8 parameters. Report time-dependent SIs rather than a single snapshot; present tornado plots at $t = 10, t_{\text{peak}}, 90$ days.

(d) Presenting results to identify intervention targets.

Produce a 2×4 panel figure: left column shows $S_{p_j}^I(t)$ curves for all 8 parameters over time (color-coded), highlighting the top 3 by $|S_{p_j}^{\cdot}|$ at t_{peak}. Right column shows tornado plots at the three key time points. Identify the two parameters with the highest SI for peak infections and annotate each with a concrete intervention (e.g., "reduce τ_I via oral rehydration therapy"; "reduce β_w via water treatment").

(e) Situation where FSE results might mislead.

If the cholera model includes a threshold: below a critical contamination level, the outbreak does not take off; above it, an epidemic occurs. Near this threshold, the SI for the contamination parameter may be extremely large (the model is near a bifurcation), suggesting it is overwhelmingly important. But this large SI is an artifact of proximity to the threshold, not a robust prediction about the effect of a realistic intervention. In this case, compute SI at several nominal parameter sets (e.g., $R_0 = 0.8$, 1.0, 1.2) and report only the SI values that are stable across this range. If the SI varies by an order of magnitude, report that result as a sensitivity of the sensitivity analysis itself.

Instructor note: A high-quality L5 answer addresses all five parts with specificity. Vague responses ("use tornado plots," "validate the code") without operational detail receive partial credit. The threshold/bifurcation concern in part (e) is the key connection to Chapter 8.

Exercise 4.8 *[Bloom L6]*

Solution.

Issue 1: Single time-point snapshot misses transient dynamics.

The model simulates an outbreak that peaks around $t = 60$ days and is essentially over by $t = 90$ days; by $t = 365$ days, the epidemic compartments are near their post-outbreak equilibrium. The SI values at $t = 365$ reflect the long-run equilibrium sensitivity, which is dominated by demographic parameters (birth rate, lifespan) and waning immunity — not the parameters that drove the acute outbreak dynamics. A single time snapshot at day 365 would show near-zero sensitivity to the transmission rate β (because the epidemic is over and $I(365) \approx 0$) but nonzero sensitivity to waning immunity (if the model has that component).

What should have been done: Compute $S_{p_j}^I(t)$ as a function of time, and report SI curves or tornado plots at clinically relevant time points (e.g., outbreak peak, $t = 30$ days, $t = 60$ days). The authors likely missed that the most influential parameters change over the course of the outbreak.

Issue 2: Reporting "a single set of values for the entire epidemic" is a category error.

FSE-computed sensitivity indices are time-dependent: $S^I_{p_j}(t)$ is a curve, not a number. Reporting one set of values implies either (a) the authors evaluated sensitivities at a single time point (addressed above), or (b) they computed a time-averaged SI, which requires specifying the weighting and is rarely the right summary. Neither is the same as "the sensitivity for the entire epidemic." A weighted average SI obscures whether a parameter is most important early, at peak, or late, and loses the information needed for timed interventions.

What should have been done: Report time-dependent SI curves, or if a scalar summary is needed, report the SI of the integral $J = \int_0^T I(t)\, dt$ (total infectious burden), which is a well-defined scalar QOI. This is exactly the QOIused in the adjoint derivation of Chapter 6.

Optional additional issue: Running the model for 365 days when the epidemic is essentially over at day 90 introduces numerical noise in the SI at late times (the SI of a nearly-zero QOIis numerically unstable). The reported "single set of values" at day 365 may be dominated by floating-point artifacts.

Instructor note: The two methodological issues are closely related: both stem from treating a time-dependent quantity (SI curve) as if it were a scalar. Award 2 points for correctly identifying the time-dependence issue with a specific alternative. Award 1 point for the category error (FSE gives curves, not numbers). The optional issue about numerical stability at $I \approx 0$ is worth bonus credit. Students who only say "should have used more time points" without identifying *why* the late-time values are misleading receive partial credit.

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5 Chapter 5: The Adjoint in Linear Algebra

Exercise 5.1 *[Bloom L2]*
Solution.

The three-step adjoint strategy in plain words:

1. Solve the forward problem to find the state. Store the solution.
2. Solve one backward (adjoint) problem for each response functional, using the matrix transposed from the forward problem.
3. For each parameter, compute the sensitivity as a dot product of the adjoint solution with the parameter's contribution vector. No additional solves are needed.

Why the adjoint equation $A^T\vec{v} = \vec{c}$ does not depend on q_j:
The adjoint equation encodes "which directions in state space affect the response $J = \vec{c}^T\vec{u}$?" This question is entirely about J (encoded in $\vec{c}$) and the forward operator (encoded in A). It does not involve any particular parameter q_j, because the adjoint does not ask what the state is — only which state directions matter for J. The parameter-dependence enters only in the dot product step ($\partial J/\partial q_j = \vec{v}^T\partial\vec{b}/\partial q_j$), which costs just one inner product per parameter once $\vec{v}$ is known.

Why this makes the adjoint efficient: With m parameters, the forward approach requires m forward solves (one FSE per parameter). The adjoint requires only 1 adjoint solve (independent of m), then m dot products. For large m, the savings are a factor of m.

Instructor note: The key conceptual point: the adjoint asks about J and A, not about specific parameters. Students often confuse the adjoint solve with a parameter-dependent computation. Full credit requires explaining why $A^T\vec{v} = \vec{c}$ has no q_j on either side.

Exercise 5.2 *[Bloom L3]*
Solution.
System: $A\vec{u} = \vec{b}$ with $A = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, $\vec{b} = (q, 2q)^T$, $J = u_1 + u_2$ ($\vec{c} = (1,1)^T$).

(a) Forward solution at $q = 1$. $A^{-1} = \frac{1}{5}\left(\begin{smallmatrix} 2 & -1 \\ -1 & 3 \end{smallmatrix}\right)$, so $\vec{u} = A^{-1}\vec{b} = \frac{1}{5}(2\cdot 1 - 1\cdot 2, -1\cdot 1 + 3\cdot 2)^T = \frac{1}{5}(0,5)^T = (0,1)^T$. Hence $u_1 = 0$, $u_2 = 1$.

(b) FSE for $\partial\vec{u}/\partial q$. Differentiate $A\vec{u} = \vec{b}$ with respect to q: $A\,(\partial\vec{u}/\partial q) = \partial\vec{b}/\partial q = (1,2)^T$. Solve: $\partial\vec{u}/\partial q = A^{-1}(1,2)^T = \frac{1}{5}(2-2, -1+6)^T = \frac{1}{5}(0,5)^T = (0,1)^T$.

(c) Adjoint equation. $A^T\vec{v} = \vec{c}$. Since A is symmetric, $A^T = A$, so $A\vec{v} = (1,1)^T$: $\vec{v} = A^{-1}(1,1)^T = \frac{1}{5}(2-1, -1+3)^T = \frac{1}{5}(1,2)^T$.

(d) Sensitivity via adjoint formula. $\partial J/\partial q = \vec{v}^T\,(\partial\vec{b}/\partial q) = \frac{1}{5}(1,2)\cdot(1,2)^T = \frac{1}{5}(1+4) = 1$.

(e) Direct verification. $J(q) = u_1(q) + u_2(q)$. From (b), $\partial J/\partial q = \vec{c}^T(\partial\vec{u}/\partial q) = (1,1)\cdot(0,1)^T = 1$. ✓

Instructor note: The system has $\vec{b} = q(1,2)^T$, so $J = u_1 + u_2$ is linear in q and $\partial J/\partial q = 1$ exactly. Students should note that both methods (FSE and adjoint) give the same answer — this is Unifying Idea 2 at work.

Exercise 5.3 *[Bloom L3]*

Solution.

The decision rule: use FSE when $m < r$; use adjoint when $r < m$. FSE requires m forward solves (plus 1 baseline); adjoint requires r adjoint solves (plus 1 forward).

(a) $m = 5$ parameters, $r = 20$ functionals. FSE: $m + 1 = 6$ solves. Adjoint: $r + 1 = 21$ solves. **Use FSE** $(6 < 21)$.

(b) $m = 100$ parameters, $r = 3$ functionals. FSE: 101 solves. Adjoint: 4 solves. **Use adjoint** $(4 \ll 101)$. Speedup factor: $\approx 25\times$.

(c) $m = 8$ parameters, $r = 8$ functionals. FSE: 9 solves. Adjoint: 9 solves. Both require the same cost. **Either works**; choose based on implementation availability. The crossover is at $m = r$.

(d) $m = 1{,}000$ parameters, $r = 1$ functional, $t = 0.1$ s/solve. FSE: $1001 \times 0.1 = 100.1$ s. Adjoint: $2 \times 0.1 = 0.2$ s. **Use adjoint.** Speedup: $100.1/0.2 \approx \mathbf{500}\times$.

Instructor note: Part (d) illustrates the regime where the adjoint is not just convenient but essential: 1000 parameters with FSE would take the same time as with any other parameter-based approach, while the adjoint reduces it to 2 solves regardless of m. This is the core motivation for Chapters 5–6.

Exercise 5.4 *[Bloom L4]*

Solution.

The symbiotic equilibrium is $\bar{u}_1 = (1 - a)/(1 - ac) = 0.625$, $\bar{u}_2 = (1 - c)/(1 - ac) = 0.75$ (from Chapter 4, Ex 4.4). The equilibrium Jacobian J_F and FSE system are as computed in Ex 4.4.

(a) FSE approach: 3 solves for 6 sensitivities.

The FSE gives $\partial\vec{u}/\partial p_j$ for each of the 3 parameters (a, b, c) by solving the same 2×2 linear system $J_F\,(\partial\vec{u}/\partial p_j) = -\partial\vec{f}/\partial p_j$. This is 3 solves, each giving $(\partial\bar{u}_1/\partial p_j,\ \partial\bar{u}_2/\partial p_j)$.

Numerical results at $a = 0.5$, $b = 2$, $c = 0.4$ (from Ex 4.4 for $p = c$; deriving for a and b similarly):

	$S_a^{\cdot}$	$S_b^{\cdot}$	$S_c^{\cdot}$
$\bar{u}_1$	-0.250	0	$+0.250$
$\bar{u}_2$	$+0.208$	0	-0.417

Derivation for $\partial/\partial a$: $-\partial\vec{f}/\partial a = (0, b\bar{u}_1\bar{u}_2/\ldots)\ldots$ but $f_1 = \bar{u}_1(1 - \bar{u}_1 - a\bar{u}_2) = 0$, so $\partial f_1/\partial a = -\bar{u}_1\bar{u}_2 = -0.625(0.75) = -0.46875$. $\partial f_2/\partial a = 0$ (only f_1 contains a). Solve: $J_F(\partial\vec{u}/\partial a) = (0.46875, 0)^T$, giving $S_a^{\bar{u}_1} = (a/\bar{u}_1)\partial\bar{u}_1/\partial a = (0.5/0.625)(-0.3125) = -0.250$.

For b: Only $f_2 = b\bar{u}_2(1 - \bar{u}_2 - c\bar{u}_1) = 0$, so $\partial f_2/\partial b = \bar{u}_2(1 - \bar{u}_2 - c\bar{u}_1) = 0$ (at equilibrium). Thus $\partial\vec{u}/\partial b = \vec{0}$: $S_b^{\bar{u}_1} = S_b^{\bar{u}_2} = 0$.

(b) Adjoint approach: 2 solves for 6 sensitivities.

Two response functionals: $J_1 = \bar{u}_1$ ($\vec{c}_1 = (1, 0)^T$) and $J_2 = \bar{u}_2$ ($\vec{c}_2 = (0, 1)^T$). Two adjoint solves: $J_F^T\vec{v}_k = \vec{c}_k$ for $k = 1, 2$. Then $\partial J_k/\partial p_j = \vec{v}_k^T(-\partial\vec{f}/\partial p_j)$ for each j.

(c) Verification: Both give the table above. ✓

(d) Comparison. FSE uses 3 solves (one per parameter), adjoint uses 2 solves (one per QOI). Adjoint is better here because $r = 2 < m = 3$. FSE would be preferred when $m < r$, e.g., for a model with 2 parameters and 5 response functionals.

Instructor note: The zero SIs for b are a natural consequence of the equilibrium structure: b scales f_2 uniformly, so at equilibrium ($f_2 = 0$), $\partial f_2/\partial b = 0$. This is a structural zero, not a numerical artifact. Students who get nonzero SI for b have an error in computing $\partial \vec{f}/\partial b$.

Exercise 5.5 *[Bloom L4]*

Solution.

$R_0 = k\beta\nu/[(\nu + \mu)(\gamma + \mu)]$.

(a) Computation graph. Intermediate nodes: $n_1 = k\beta$, $n_2 = n_1\nu$, $n_3 = \nu + \mu$, $n_4 = \gamma + \mu$, $n_5 = n_3 n_4$, $R_0 = n_2/n_5$.

(b) Forward mode for $\partial R_0/\partial\nu$. Propagate $\dot{\nu} = 1$ (seeding the derivative w.r.t. ν): $\dot{n}_1 = 0$, $\dot{n}_2 = n_1 \cdot 1 + \nu \cdot 0 = k\beta$, $\dot{n}_3 = 1$, $\dot{n}_4 = 0$, $\dot{n}_5 = \dot{n}_3 n_4 + n_3 \dot{n}_4 = (\gamma + \mu)$. $\partial R_0/\partial\nu = (\dot{n}_2 n_5 - n_2 \dot{n}_5)/n_5^2 = (k\beta(\nu + \mu)(\gamma + \mu) - k\beta\nu(\gamma + \mu))/n_5^2 = k\beta(\gamma + \mu)\mu/[(\nu + \mu)^2(\gamma + \mu)^2] = k\beta\mu/[(\nu + \mu)^2(\gamma + \mu)]$. Arithmetic operations: approximately 8 multiplications, 4 additions.

(c) Reverse mode for all 5 partial derivatives. Seed $\bar{R}_0 = 1$. Backpropagate: $\bar{n}_2 = \bar{R}_0/n_5 = 1/n_5$, $\bar{n}_5 = -\bar{R}_0 n_2/n_5^2 = -R_0/n_5$. $\bar{n}_3 = \bar{n}_5 n_4$, $\bar{n}_4 = \bar{n}_5 n_3$. $\bar{n}_1 = \bar{n}_2\nu$, $\bar{\nu} = \bar{n}_2 n_1 + \bar{n}_3$. $\bar{k} = \bar{n}_1\beta$, $\bar{\beta} = \bar{n}_1 k$. $\bar{\gamma} = \bar{n}_4$, $\bar{\mu} = \bar{n}_3 + \bar{n}_4$. One backward pass gives all 5 derivatives simultaneously.

(d) Normalized SI tornado plot. Using interpretable parameters $\tau_E = 1/\nu$, $\tau_I = 1/\gamma$, $L = 1/\mu$: $S_k^{R_0} = 1$, $S_\beta^{R_0} = 1$, $S_{\tau_E}^{R_0} = -\tau_E/(\tau_E + L)$, $S_{\tau_I}^{R_0} = \tau_I L/(\tau_I + L)\cdot(1/R_0\tau_I) \approx L/(L + \tau_I)$, $S_L^{R_0} \approx 0$ (for $L \gg \tau_E, \tau_I$). The tornado plot shows k and β dominant (SI = 1), with τ_E and τ_I small for $L = 10{,}000$ days.

Instructor note: The reverse-mode computation (part c) is exactly the adjoint method applied to a scalar computation graph. This exercise bridges the algebraic adjoint (Chapter 5) with automatic differentiation (Chapter 7).

Exercise 5.6 *[Bloom L5]*

Solution.

Adjoint SA design for river water-quality model: $m = 40$, $r = 2$.

(a) Form of the adjoint equation. The river-quality model is a system of ODEs or PDEs. For a generic ODE model $\dot{\vec{u}} = \vec{F}(\vec{u}; \vec{p})$ with response $J_k = \int_0^T g_k(\vec{u})\,dt + h_k(\vec{u}(T))$, the adjoint ODE for response k is:

$$-\dot{\vec{\lambda}}^{(k)} = J_F^T(\vec{u}(t))\,\vec{\lambda}^{(k)} - \nabla_{\vec{u}} g_k(\vec{u}(t)), \qquad \vec{\lambda}^{(k)}(T) = \nabla_{\vec{u}} h_k(\vec{u}(T)).$$

Two adjoint equations (one per QOI), both depending on the stored forward solution $\vec{u}(t)$.

(b) Number of adjoint solves: 2. One adjoint solve per response functional. Each solve runs backward from $t = T$ to $t = 0$, giving $\vec{\lambda}^{(k)}(t)$. Total cost: 1 forward solve (store) + 2 adjoint solves. Compare with FSE: 1 + 40 forward solves. The adjoint is $\approx 20\times$ cheaper.

(c) Validation. Verify that the adjoint-computed $\partial J_k/\partial p_j$ matches centered finite differences for all 40 parameters: use $h = 6 \times 10^{-6}|p_j^*|$ and check agreement to 5 significant figures. Also verify the adjoint at a simpler test case with analytic sensitivity (e.g., a single-compartment degradation model where J and $\partial J/\partial p$ are known analytically).

(d) Advising regulators. Produce two tornado plots (one per QOI) showing all 40 parameters sorted by $|\partial J_k/\partial p_j|$. Annotate each parameter with its plain-language interpretation and the regulatory lever available: "Reducing phosphorus loading at source i by 10% reduces downstream concentration by approximately 2.3%." Identify the top 5 sources for each QOIand report whether the same parameters appear in both tornado plots (co-optimization opportunity) or different ones (trade-off between the two monitoring objectives).

(e) Potential misleading results. If the river model has a threshold effect (e.g., dissolved oxygen below a critical level triggers biological collapse), the adjoint — which linearizes around the current state — may show small SI values for parameters that would be critical if the threshold were crossed. To detect: run the model at several nominal parameter sets spanning the plausible range; if any set pushes the model near the threshold, the adjoint at the current operating point is not representative. Correction: report sensitivity results at multiple operating points and flag which parameters bring the system close to critical thresholds.

Instructor note: Full credit requires all five components with operational specificity. The validation plan (part c) must include both a numerical check (FD agreement) and an analytic check (simple test case). Vague statements like "validate the code" are insufficient.

Exercise 5.7 *[Bloom L6]*

Solution.

(a) Is the first sentence correct?

Yes, with an important qualifier. Both FSE and adjoint compute $\partial J/\partial p_j$ for all j, and the mathematical result is identical: $\partial J/\partial p_j = \vec{v}^T \partial\vec{b}/\partial p_j$ (adjoint) $= \vec{c}^T(\partial\vec{u}/\partial p_j)$ (FSE), and these two expressions are equal because $\vec{c}^T A^{-1} = \vec{v}^T$ (by definition of the adjoint). So the *answer* is identical. The adjoint is faster when $m \gg r$; if $m < r$, the FSE is actually faster. "Just faster" is only correct in the typical regime $m \gg r$.

(b) Is the second sentence correct?

No, for at least two reasons.

Situation 1: The number of parameters grows. If the model is extended from $m = 10$ to $m = 200$ parameters (e.g., adding spatially distributed source terms), the FSE cost grows as 200 solves while the adjoint remains at $r + 1$ solves. At $m = 200$ and $r = 1$, switching from FSE to adjoint gives a 100× speedup.

Situation 2: Multiple response functionals for a well-optimized adjoint. If a colleague needs to compute sensitivities for an additional 5 QOIson the same model, the FSE cost is unchanged (still $m + 1$ solves) while the adjoint adds only 5 more backward solves. For models where solving the adjoint ODE is cheap compared to evaluating $\partial\vec{f}/\partial p_j$ for all j, the adjoint is the only viable option at large m.

(c) When is FSE preferred?

FSE is preferred when $m < r$: the number of parameters is less than the number of response functionals. FSE costs $m + 1$ solves; adjoint costs $r + 1$ solves.

Specific example where FSE is strictly better: $m = 3$ parameters (e.g., a simple SIR model with k, β, τ), $r = 50$ response functionals (sensitivity at 50 time points, or for 50 spatial locations). FSE: 4 solves. Adjoint: 51 solves. FSE is $51/4 \approx 13\times$ faster.

Instructor note: The key insight is that "adjoint is always better" is a myth. The correct statement is: use the approach whose cost ($\min(m, r) + 1$ solves) is smaller. This is Table 5.1 in the text. Students who say only "adjoint is better when m is large" miss the symmetric counterexample. Full credit requires a specific numerical example in part (c).

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6 Chapter 6: The Adjoint ODE

Exercise 6.1 *[Bloom L3]*

Solution.

Logistic ODE: $\dot{u} = ru(1 - u/K)$, $J = \int_0^T u\,dt$. Here $F(u; r, K) = ru(1 - u/K)$, $g(u) = u$, $h = 0$.

(a) Adjoint ODE and terminal condition.

The running cost gradient: $\partial g/\partial u = 1$. The state Jacobian: $F_u = r(1 - 2u/K)$. Adjoint ODE (backward):

$$-\dot{\lambda} = F_u\,\lambda - \frac{\partial g}{\partial u} = r\left(1 - \frac{2u(t)}{K}\right)\lambda - 1, \quad \lambda(T) = 0.$$

Equivalently (standard sign for integration backward with `ode45`):

$$\dot{\lambda} = 1 - r\left(1 - \frac{2u(t)}{K}\right)\lambda.$$

(b) Linearity in λ.

The adjoint ODE is linear in λ: $\dot{\lambda} = 1 - r(1 - 2u(t)/K)\lambda$. The coefficient of λ is $-r(1 - 2u(t)/K)$, which depends on $u(t)$ but not on λ itself. The forcing term is $+1$. This is a first-order linear ODE in λ, even though the forward logistic ODE is nonlinear in u. This linearity is guaranteed by the Lagrange multiplier derivation (Theorem 6.1): the adjoint is always the adjoint of the *linearized* forward equation.

(c) Why the adjoint depends on $u(t)$.

The adjoint ODE measures how sensitive J is to perturbations of the state at time t. This sensitivity depends on how the forward dynamics amplify or dampen perturbations between time t and T, which in turn depends on $u(t)$ through the coefficient $F_u = r(1 - 2u/K)$. Near the carrying capacity ($u \approx K$), $F_u \approx -r < 0$: perturbations decay. Near $u = 0$, $F_u \approx r > 0$: perturbations grow. The adjoint must account for this varying amplification.

(d) MATLAB implementation.

```
r=0.5; K=100; u0=10; T=20;
ode_opts = odeset('RelTol',1e-8,'AbsTol',1e-10,'MaxStep',0.1);

% Forward solve
f_fwd = @(t,u) r*u*(1-u/K);
[t_fwd, U] = ode45(f_fwd, [0,T], u0, ode_opts);
u_fn = griddedInterpolant(t_fwd, U, 'pchip');

% Backward adjoint solve (tspan=[T,0])
f_adj = @(t,lam) 1 - r*(1-2*u_fn(t)/K)*lam;
[t_adj, LAM] = ode45(f_adj, [T,0], 0, ode_opts);

% Sensitivity integrals
SI_r = (r/trapz(t_fwd,U)) * trapz(t_fwd, LAM_interp.*U.*(1-U/K));
```

At nominal values, $\lambda(t)$ is large near $t = 0$ (early population growth is more influential for the integral $J = \int u\,dt$) and decreases toward $t = T$ as the population saturates at K.

Instructor note: Students often get the sign of the adjoint ODE wrong. The forward-time form is $\dot{\lambda} = 1 - r(1-2u/K)\lambda$; calling `ode45` with `tspan = [T, 0]` automatically handles backward integration. Students who negate the RHS AND use backward tspan are double-counting the sign. The key test: at T, $\lambda(T) = 0$; at $t = 0$, $\lambda(0)$ should be large and positive.

Exercise 6.2 *[Bloom L3]*

Solution.

What integration by parts accomplishes:

Step 3 transfers the time derivative d/dt from the sensitivity variable $w(t)$ (which we want to eliminate) onto the adjoint variable $\lambda(t)$ (which we can freely define). This is precisely what the matrix transpose accomplishes algebraically: if $A\vec{w} = \vec{f}$ is the FSE, then integrating $\vec{\lambda}^T(A\vec{w})$ by parts shifts the differential operator from $\vec{w}$ to $\vec{\lambda}$, resulting in $(-\dot{\vec{\lambda}} - A^T\vec{\lambda})$ multiplying $\vec{w}$. Setting this coefficient to zero defines the adjoint ODE, making the $\vec{w}$-integral vanish.

Analogy with transposition:

In the algebraic setting (Chapter 5), the FSE is $A\vec{w} = \vec{f}$ and the adjoint is $A^T\vec{v} = \vec{c}$. The sensitivity formula comes from: $\vec{c}^T\vec{w} = (A^T\vec{v})^T\vec{w} = \vec{v}^T(A\vec{w}) = \vec{v}^T\vec{f}$. The transpose moves the operator from $\vec{w}$ to $\vec{v}$. Integration by parts does the same for the differential operator: $\int \lambda\,\dot{w}\,dt = [\lambda w]_0^T - \int \dot{\lambda}\,w\,dt$ moves the time derivative from w to λ. The ODE adjoint is the continuous analog of the matrix transpose.

Direct verification of equation (6.X):

Let $p(t) = \lambda(t)w(t)$. Differentiate by product rule: $\dot{p} = \dot{\lambda}w + \lambda\dot{w}$. Integrate over $[0, T]$:

$$\int_0^T (\dot{\lambda}w + \lambda\dot{w})\,dt = [p(t)]_0^T = \lambda(T)w(T) - \lambda(0)w(0).$$

Rearranging:

$$\int_0^T \lambda\dot{w}\,dt = \lambda(T)w(T) - \lambda(0)w(0) - \int_0^T \dot{\lambda}w\,dt. \checkmark$$

This is equation (6.X) (the integration-by-parts identity for ODE adjoints).

Instructor note: The key pedagogical connection is: transpose (Chapter 5) and integration by parts (Chapter 6) are both instances of Unifying Idea 2 — the adjoint is the transpose in whatever space we are working in. In finite dimensions, transpose reverses the pairing; in function space, integration by parts reverses the temporal derivative.

Exercise 6.3 *[Bloom L4]*

Solution.

Using `run_sir_adjoint(sir_nominal())` from the repository, with $J = \int_0^{90} I\,dt$.

(a) Adjoint-computed SI values (nominal parameters).

Parameter	$\partial J/\partial p_j$ (raw)	$S^J_{p_j}$
k	$\sim +120$	$\approx +0.57$
β	$\sim +2000$	$\approx +0.57$
τ	$\sim +17$	$\approx +0.38$
L	~ -0.4	≈ -0.03

(Exact values depend on ODE solver tolerances; ratios are robust.)

(b) Why $S^J_k = S^J_\beta$.

In the SIR model, k and β appear only in the product $k\beta$. Any perturbation to k has the identical structural effect on the ODE right-hand side as a proportional perturbation to β. Since the model is invariant to the labeling $k \leftrightarrow \beta$ (they are not separately identifiable from outbreak data alone), their normalized SIs must be identical. Mathematically: $\partial F/\partial k = \beta SI/N$ and $\partial F/\partial \beta = kSI/N = (k/\beta)(\partial F/\partial k)$, so after normalization the sensitivities cancel out and become equal.

(c) Optimal intervention.

The tornado plot shows k and β tied for largest SI, followed by τ. If one intervention reduces a single parameter by 20%, the reduction in J (total infectious burden) is approximately $20\% \times |S^J_{p_j}|$ for each: k or β: $\approx 11.4\%$; τ: $\approx 7.6\%$; L: negligible. **Target k or β** (whichever is more controllable). In practice, k (contact rate) is reduced by social distancing; β (transmission probability per contact) by mask-wearing or hygiene. Both are equally effective at reducing total disease burden.

(d) Cost comparison with FSE.

FSE for the time-dependent SI $S^I_{p_j}(t)$: one augmented ODE solve ($4+4\times 4 = 20$ equations). Adjoint: one forward solve + one adjoint solve (each with ~ 4 equations). For this small model they are comparable; the adjoint becomes much faster for models with $m \gg r$.

Instructor note: The equality $S^J_k = S^J_\beta$ is a structural property of the SIR model that students should identify and explain, not just report. This connects to the identifiability material from Chapter 3: k and β are not separately identifiable from $I(t)$ data.

Exercise 6.4 *[Bloom L4]*

Solution.

FSE cost: $m+1$ ODE solves. Adjoint cost: $r+1=2$ ODE solves (1 forward + 1 adjoint for $r=1$ QOI). Each solve: 30 seconds.

m	FSE cost (solves)	Adjoint cost (solves)	Speedup
4	5	2	2.5×
10	11	2	5.5×
50	51	2	25.5×
500	501	2	250.5×

At what m is adjoint $> 10\times$ faster?

Adjoint time: $2 \times 30 = 60$ s. FSE time: $(m+1) \times 30$ s. Speedup $> 10\times$ when $(m+1)/2 > 10$, i.e., $m > 19$. At $m = 20$: FSE = 630 s, adjoint = 60 s, speedup = 10.5×. The adjoint becomes $> 10\times$ faster at $\mathbf{m \geq 20}$ **parameters.**

Instructor note: Many students write "the adjoint becomes better at $m \geq 2$" (technically true since adjoint costs 2 and FSE costs 3 at $m = 2$) but miss the practical threshold. The question asks for $> 10\times$; students should set up and solve the inequality. Note that for $r = 1$ (one QOI), the adjoint is always better than FSE for $m > 1$ — the interesting threshold for practioners is when the speedup is large enough to justify reimplementation.

Exercise 6.5 *[Bloom L4]*

Solution.

(a) Why integrate backward from T?

The adjoint variable $\lambda(t)$ answers the question: "If the state at time t were perturbed, how much would J change?" This question is inherently backward in time — the effect of a perturbation at time t propagates forward to T, and we need to know what happened between t and T to answer it. At T itself, the effect on J of a perturbation to $u(T)$ is determined directly by how J depends on the final state. Working backward from this known terminal value, we accumulate the influence of perturbations at each earlier time. Forward integration from $t = 0$ would require knowing the effect of future perturbations before we have computed them.

(b) Why $\lambda_I(t)$ is larger near $t = 0$?

$\lambda_I(t)$ measures how sensitive $J = \int I\,dt$ is to a perturbation of I at time t. A perturbation at $t = 5$ days has 85 days to propagate through the epidemic dynamics and affect the total infectious burden. A perturbation at $t = 85$ days has only 5 days left; most of the integral J has already been accumulated. So early perturbations matter more, and $\lambda_I(5) > \lambda_I(85)$.

(c) Early intervention recommendation.

From the adjoint solution at nominal parameters, $\lambda_I(5) \approx L_{\max}$ while $\lambda_I(80) \approx 0.1\,L_{\max}$. A 10% reduction in the contact rate k applied only during days 0–10 reduces J by approximately $10\% \times |S_k^J| \times$(effective weight at $t = 5$), which is proportionally larger than the same reduction applied during days 80–90. The practical recommendation: interventions (school closures, event cancellations) are most effective when implemented in the first two weeks of an outbreak.

(d) Terminal condition vs. reverse-time forward.

Running the forward ODE backward from $\vec{u}(T)$ with initial condition $\vec{0}$ would give a solution to $\dot{\vec{z}} = -\vec{F}(\vec{z})$, which is the time-reversal of the epidemic — unphysical and unrelated to the adjoint. The adjoint terminal condition $\vec{\lambda}(T) = \nabla_{\vec{u}} h(\vec{u}(T))$ encodes the sensitivity of J to the final state, which is determined by the structure of the response functional, not by the forward dynamics. For $J = \int I\,dt$ with no terminal cost, $\vec{\lambda}(T) = \vec{0}$ because J does not directly depend on the final state; the adjoint ODE then builds up the historical influence backward through the epidemic.

Instructor note: Parts (a) and (d) are the most conceptually challenging in the chapter. Common error for (d): students say both approaches "start from zero" and cannot distinguish them. The key is that the adjoint ODE coefficients depend on the *forward* solution $u(t)$, which must be stored and accessed backward. Simply integrating the forward ODE backward yields a physically different trajectory.

Exercise 6.6 *[Bloom L5]*

Solution.

SEIR model with $\vec{p} = (k, \beta, \tau_E, \tau_I, L)$, $J = \int_0^T I(t)\,dt$.

(a) 4×4 Jacobian J_F.

Let $\gamma = 1/\tau_I$, $\nu = 1/\tau_E$, $\mu = 1/L$.

$$J_F = \frac{\partial(F_S, F_E, F_I, F_R)}{\partial(S, E, I, R)} = \begin{pmatrix} -\lambda-\mu & 0 & -k\beta S/N & 0 \\ \lambda & -(\nu+\mu) & k\beta S/N - k\beta I/N & 0 \\ 0 & \nu & -(\gamma+\mu) & 0 \\ 0 & 0 & \gamma & -\mu \end{pmatrix}$$

where $\lambda = k\beta I/N$ (force of infection). Off-diagonal terms in column 3 arise because I appears in λ.

(b) Four adjoint ODEs and terminal conditions.

For $J = \int I\,dt$: $\partial g/\partial\vec{u} = (0, 0, 1, 0)^T$, $h = 0$. Terminal conditions: $\vec{\lambda}(T) = \vec{0}$. Adjoint ODEs:

$$\begin{aligned} -\dot{\lambda}_S &= (-\lambda-\mu)\lambda_S + \lambda\,\lambda_E \\ -\dot{\lambda}_E &= -(\nu+\mu)\lambda_E + \nu\lambda_I \\ -\dot{\lambda}_I &= -\frac{k\beta S}{N}\lambda_S + \frac{k\beta S}{N}\lambda_E - (\gamma+\mu)\lambda_I + \gamma\lambda_R - 1 \\ -\dot{\lambda}_R &= -\mu\lambda_R \end{aligned}$$

(c) Sensitivity formulas.

$dJ/dp_j = -\int_0^T \vec{\lambda}^T(\partial\vec{F}/\partial p_j)\,dt$. Computing $\partial\vec{F}/\partial p_j$:

$dJ/dk = -\int_0^T(-\lambda_S + \lambda_E)(p.beta \cdot SI/N)\,dt$. $dJ/d\beta$: identical form with k and β exchanged.
$dJ/d\tau_E = -\int_0^T \lambda_I(E/\tau_E^2)\,dt$ (from $\partial\nu/\partial\tau_E = -1/\tau_E^2$, $\partial F_E/\partial\tau_E = -E/\tau_E^2$, $\partial F_I/\partial\tau_E = E/\tau_E^2$).
$dJ/d\tau_I = -\int_0^T \lambda_I(I/\tau_I^2) - \lambda_R(-I/\tau_I^2)\,dt$. dJ/dL: similar via $\partial\mu/\partial L = -1/L^2$.

(d) MATLAB implementation strategy. Forward solve: 4-equation SEIR ODE, save dense output. Build interpolants for all 4 state variables. Backward solve: 4-equation adjoint ODE, using stored interpolants for $S(t), E(t), I(t), R(t)$. Compute integrals via `trapz` on forward time grid.

(e) Additional cost vs. SIR adjoint. The SEIR adjoint requires 4 equations vs. 3 for SIR (one additional adjoint variable λ_E). Storage doubles from $\sim 3 \times N_t$ to $\sim 4 \times N_t$ values. The computational cost is essentially identical: one additional ODE component adds negligible time to the adjoint solve. The main additional cost is writing and verifying the 4×4 Jacobian and 5 forcing terms.

Instructor note: Students who write J_F^T and then compute the adjoint ODE by substituting into the formula $-\dot{\vec{\lambda}} = J_F^T\vec{\lambda} - \nabla_{\vec{u}}g$ should get exactly the four equations in part (b). Common error: forgetting that J_F is evaluated along the forward solution $\vec{u}(t)$, not at a fixed point.

Exercise 6.7 *[Bloom L6]*

Solution.

Three causes of a 15% discrepancy between adjoint and FD, without a coding error:

Cause 1: Inconsistent step sizes in the forward/backward solvers.

The adjoint ODE depends on the stored forward solution $\vec{u}(t)$ through interpolation. If the forward ODE is solved with a loose tolerance (e.g., `RelTol = 1e-3`) but the adjoint is solved tightly, the interpolated $\vec{u}(t)$ values contain errors proportional to the forward tolerance, which propagate into the adjoint sensitivity.

Diagnostic: Tighten the forward solver tolerance by 10^2 and recompute. If the adjoint sensitivities change by more than 1%, forward solver error is the cause.

Correction: Use tight tolerances ($\leq 10^{-8}$) for both forward and adjoint solves. Consider a fixed-step integrator if adaptive stepping causes inconsistency.

Cause 2: Poor interpolation of the forward solution.

The adjoint ODE is integrated backward and must evaluate $\vec{u}(t)$ at times not necessarily on the forward solver's output grid. Linear interpolation (the default in some implementations) introduces $O(h^2)$ errors in the adjoint coefficients; for stiff models with rapidly varying states, this can cause $> 10\%$ errors in sensitivities.

Diagnostic: Compare adjoint sensitivities using linear, cubic (PCHIP), and quintic interpolation of the stored forward solution. If results change by more than 1%, interpolation order is the cause.

Correction: Use a high-order dense output method (`pchip` or `griddedInterpolant` with 'spline') and store the forward solution at a fine time grid.

Cause 3: The model is nonlinear enough that the linearization underlying the adjoint is inaccurate.

The adjoint computes the *derivative* of J with respect to p_j at the nominal parameter point. If the model is highly nonlinear, the finite-difference estimate $[J(p_j + h) - J(p_j - h)]/(2h)$ includes higher-order terms that the adjoint (which is exact for the linearized model) ignores.

Diagnostic: Compare FD estimates at several step sizes h and verify they converge as $h \to 0$ (ruling out truncation error in FD). If FD values are consistent across step sizes but differ from the adjoint, nonlinearity is the cause.

Correction: For parameters where the adjoint and FD disagree by $> 5\%$, report both and note that the adjoint provides the exact infinitesimal sensitivity while FD captures the finite-perturbation effect. Consider reporting the FD estimate for policy, since it reflects the actual change from a realistic intervention.

Instructor note: A 15% discrepancy between adjoint and FD is large enough to be concerning but not obviously a coding error. The three causes above cover the most common scenarios. Students who only identify "step-size error" as a single cause receive partial credit; the distinction between forward solver tolerance (Cause 1), interpolation accuracy (Cause 2), and model nonlinearity (Cause 3) is essential for correct diagnosis. Note that all three causes can coexist; the diagnostic tests are designed to isolate each one.

7 Chapter 7: Algorithmic Differentiation

Exercise 7.1 *[Bloom L3]*

Solution.

Annuity function: $A = P\,r(1+r)^n/[(1+r)^n - 1]$ at $P = 1000$, $r = 0.05$, $n = 20$.

(a) Numerical value. $(1.05)^{20} = 2.6533$. $A = 1000 \times 0.05(2.6533)/(2.6533 - 1) = 1000 \times 0.13277/1.6533 = \mathbf{80.24}$ per period.

(b) Analytic derivatives. Let $R = (1+r)^n$. Then $A = PrR/(R-1)$.

$$\frac{\partial A}{\partial P} = \frac{rR}{R-1} = A/P = 0.08024. \quad S_P^A = \frac{P}{A} \cdot \frac{A}{P} = 1.$$

$$\frac{\partial A}{\partial r} = P\frac{(R-1)(R + rnR/r) - rR \cdot nR/r}{(R-1)^2} \text{ [chain rule; simplify to get } S_r^A\text{].}$$

Using log-derivative for S_r^A: $\ln A = \ln P + \ln r + n\ln(1+r) - \ln[(1+r)^n - 1]$. $S_r^A = \frac{d\ln A}{d\ln r} = 1 + \frac{nr}{1+r} - \frac{nr(1+r)^{n-1}}{(1+r)^n - 1} \cdot r$. At $r = 0.05$, $n = 20$: $S_r^A \approx 1 + 0.9524 - 0.9524 \times 2.6533/1.6533 \approx 1 + 0.9524 - 1.529 \approx \mathbf{0.424}$.

$S_n^A = \frac{n}{A} \cdot \frac{\partial A}{\partial n} = r\ln(1+r) \cdot n \cdot \frac{(R-1)-R}{(R-1)^2/(R-1)} \cdots$ Numerically: $S_n^A \approx -\mathbf{0.468}$ (payments increase as n decreases).

(c) Forward mode for $\partial A/\partial r$ (seeding $\dot{r} = 1$). $\dot{R} = n(1+r)^{n-1} = 20(1.05)^{19} = 50.539$. $\dot{A} = P[\dot{r}R + r\dot{R})(R-1) - rR\dot{R}]/(R-1)^2$. Numerically: matches analytic result $\partial A/\partial r \approx 680.2$. $S_r^A = (r/A) \times 680.2 = (0.05/80.24) \times 680.2 \approx 0.424$. ✓

(d) Reverse mode (all three in one pass). Seed $\bar{A} = 1$. Backpropagate: $\bar{R} = P[r(R-1) - rR]/(R-1)^2$, then $\bar{r} = PR/(R-1) + \bar{R} \cdot n(1+r)^{n-1}$, $\bar{P} = rR/(R-1)$, $\bar{n} = \bar{R} \cdot R\ln(1+r)$. One backward pass yields all three. ✓

(e) Operation counts. Forward mode (one parameter): $\sim$8 multiplications, 3 additions. Reverse mode (all 3): $\sim$12 multiplications, 6 additions. Forward-mode for all 3 parameters: $3 \times 8 = 24$ operations. Reverse mode is $24/12 = 2\times$ cheaper for all parameters combined.

Instructor note: Part (b): the analytic S_r^A via the log-derivative is the cleanest approach. Students who expand via the quotient rule often make algebra errors. Encourage log-differentiation for compound expressions involving products and ratios.

Exercise 7.2 *[Bloom L4]*

Solution.

(a) Climate model, 800 parameters, 2 response functionals, 10 h/run. Use adjoint AD. With $m = 800$ and $r = 2$, FSE costs 801 runs ($\approx$ 8010 hours) while adjoint costs 3 runs (30 hours). If the model code supports AD (e.g., via Tapenade for Fortran), implement the adjoint. If not, central-difference FSE is not feasible; use adjoint or restrict to Morris screening ($\approx$ 8000 evaluations = 80,000 hours — still infeasible). This is a case where developing the adjoint is the only practical option.

(b) PK model, 5 parameters, 8 time points, analytic solutions. Use analytic FSE (or log-derivatives directly). With analytic solutions, $\partial u/\partial p_j$ can be derived symbolically.

No AD or numerical FD is needed. Cost: trivial. Using MATLAB `diff` or SymPy once gives all 5×8 sensitivities.

(c) Traffic simulation, 40 parameters, compiled C++, 3 min/run. Use centered finite differences. AD is unavailable (compiled black box). Centered FD costs 81 runs ×3 min = 243 min ≈ 4 hours. Feasible overnight. Adjoint would require modifying the C++ source (possible but costly in developer time). Morris screening (15 trajectories, ≈ 615 runs) is also feasible if nonlinearity is a concern.

(d) MATLAB ODE45 model, 10 parameters, 1 qoi, `if` statement. Use centered finite differences. The `if` statement makes the model non-differentiable at the quarantine threshold; forward-mode AD and the adjoint will both fail at parameter values that trigger the discontinuity. Centered differences detect the discontinuity as a large error indicator (the left and right estimates diverge near threshold). Cost: 21 runs. Use the U-curve diagnostic to identify non-smooth parameters.

Instructor note: Part (d) is the key discriminating item: students who recommend AD or adjoint without noting the `if`-statement issue have not read the question carefully. Chapter 8 covers this exact failure mode. The correct approach for non-smooth models is FD (which still works, though less accurately near the threshold) with careful step-size checking.

Exercise 7.3 *[Bloom L4]*

Solution.

(a) dlarray for $R_0(k, \beta, \tau) = k\beta\tau$.

```
k = dlarray(5); beta = dlarray(0.06); tau = dlarray(7);
R0 = k * beta * tau;
[dR0_dk, dR0_dbeta, dR0_dtau] = dlgradient(R0, k, beta, tau);
% dR0_dk = beta*tau = 0.42; dR0_dbeta = k*tau = 35; dR0_dtau = k*beta = 0.3
```

(b) Verification. $\partial R_0/\partial k = \beta\tau = 0.42$ ✓; $\partial R_0/\partial\beta = k\tau = 35$ ✓; $\partial R_0/\partial\tau = k\beta = 0.3$ ✓. Normalized: $S_k^{R_0} = (k/R_0) \cdot 0.42 = (5/2.1) \cdot 0.42 = 1$ ✓.

(c) SIR endemic equilibrium. At the endemic equilibrium $\bar{I} > 0$: numerically solve $\vec{f}(\vec{u}; \beta) = \vec{0}$ for $\bar{I}(\beta)$. Since `ode45` is not natively differentiable via `dlarray`, implement a Newton iteration inside a `dlarray` context:

```
beta_dl = dlarray(0.06);
% Newton iteration (differentiable):
u = dlarray([999; 1; 0]);  % initial guess
for k = 1:20
    F = sir_rhs_dl(u, beta_dl);  % model RHS as dlarray function
    J = sir_jac_dl(u, beta_dl);
    u = u - J \ F;
end
dIbar_dbeta = dlgradient(u(2), beta_dl);
```

Compare with FSE result from Chapter 4 ($S_\beta^{\bar{I}}$ at equilibrium).

(d) Applying dlgradient through ode45. `ode45` is not a differentiable MATLAB function: it uses internal adaptive stepping, logical branching, and memory allocation that `dlarray` cannot trace through the computation graph. Calling `dlgradient` after `ode45` will produce an error or silently return zero gradients. **Remedy:** Either (1) implement the ODE solve as a differentiable fixed-step method (e.g., explicit Euler or RK4 loop in `dlarray`), or (2) use the adjoint ODE approach (Chapter 6), which bypasses the need to differentiate through the ODE solver itself.

Instructor note: Part (d) is the central limitation of black-box AD applied to ODE solvers. The forward-mode approach (seeding $\dot{\beta} = 1$ through the ODE RHS) works only for fixed-step explicit integrators. The adjoint approach of Chapter 6 circumvents this entirely.

Exercise 7.4 *[Bloom L5]*

Solution.

Dengue model SA design: 8 state variables, 14 parameters, 2 QOIs, 2 min/run.

(a) Method: analytic adjoint ODE. With $m = 14$ and $r = 2$, FSE costs 15 runs (30 min) while adjoint costs 3 runs (6 min). For a published dengue model with known equations, deriving the 8×8 Jacobian analytically is feasible (~ 1 day of derivation). Adjoint is preferred for efficiency and for providing exact sensitivities without step-size issues. If derivation time is a constraint, centered FD at 29 runs (58 min) is also acceptable.

(b) Computation time. Adjoint: 1 forward + 1 adjoint (for $r = 2$, need 2 adjoint solves) $= 3$ runs $\times$ 2 min $=$ **6 minutes.** Centered FD: 29 runs $\times$ 2 min $=$ **58 minutes.**

(c) Validation. Verify $S_k^{R_0} = S_{\beta_h}^{R_0} = 1$ analytically against adjoint output. Compare adjoint sensitivities with centered FD for all 14 parameters; require agreement to 4 significant figures. Test the adjoint code on a single-compartment analytic submodel with known sensitivities.

(d) Handling structural correlation between σ_v and σ_h. Report standard (marginal) SIs and total-derivative SIs. The coupling coefficient is estimated from the epidemiological literature or from field measurements of the joint distribution of biting rates. If σ_v increases by $x\%$, σ_h increases by $c_{vh} \cdot x\%$. The total-derivative SI of each QOI with respect to σ_v accounts for this induced change in σ_h. Report both sets side-by-side and flag the structural correlation explicitly in the methods section.

(e) Presentation for public health audience. Two tornado plots (one per QOI), with parameters labeled by plain-language intervention descriptions: "Reducing average mosquito biting rate by 10% reduces disease prevalence by approximately X%." Use a 3-color coding: green (large positive SI — increasing this parameter helps reduce burden), red (large negative SI — decreasing this parameter reduces burden), gray (small SI — negligible influence). Limit to top 5 parameters per QOI for clarity.

Instructor note: This is a realistic applied SA design exercise. Full credit requires all five components with specificity. The structural correlation handling (d) is the most commonly missed; students who only say "account for correlation" without specifying how receive partial credit.

Exercise 7.5 *[Bloom L6]*

Solution.

The most likely technical cause: Incorrect treatment of the adaptive step-size logic during reverse-mode AD of the ODE solver.

(a) Root cause. Adaptive-step-size ODE solvers (like `ode45`) use internal comparisons and branching to decide step sizes. When reverse-mode AD is applied through the ODE solver, the branching decisions made during the forward pass are replayed during the backward pass. For parameters that appear in the mortality rate expressions (which affect stability and thus step-size selection), a perturbation can change which branches are taken, making the reverse-mode computation inconsistent with the actual forward computation. This manifests as 10–20% errors that are largest for parameters affecting stability (mortality rates).

(b) Two diagnostic tests.

Test 1: Fixed-step integrator comparison. Re-implement the ODE solve with a fixed-step explicit RK4 method. Apply reverse-mode AD to this fixed-step implementation. If the discrepancy disappears, the adaptive step-size branching is the cause.

Test 2: Finite-difference consistency. For the discrepant parameters, compute FD sensitivities at $h = 10^{-4}$ and $h = 10^{-5}$. If these agree with each other but not with the AD result, the AD implementation is wrong; if they also disagree, the model may be genuinely discontinuous near the nominal parameter values.

(c) Two remedies and trade-offs.

Remedy 1: Fixed-step AD-compatible integrator. Replace the adaptive solver with a fixed-step RK4 inside the AD tape. *Trade-off:* Fixed-step integrators require choosing a step size that ensures accuracy for all parameter values; for stiff models, this may require very small steps and high computational cost. Advantage: mathematically clean, gradients are exact for the fixed-step approximation.

Remedy 2: Discrete adjoint (checkpointing). Use the adjoint ODE approach (Chapter 6) with the adaptive forward solver stored at checkpoints. The adjoint ODE is integrated backward using the stored checkpoints, avoiding the need to differentiate through the ODE solver's branching logic. *Trade-off:* Requires implementing the adjoint ODE separately (additional derivation and coding); storage grows with the number of checkpoints. Advantage: handles adaptive steps naturally and is more memory-efficient.

(d) When to conclude genuine SA result.

If the FD sensitivities computed at two consistent step sizes (h and $h/2$) agree with each other within 1% but differ from the AD result by 10–20%, the AD implementation is suspect. However, if FD estimates also show high variability (differing by $> 5\%$ as h is halved), this suggests the model is genuinely non-smooth and the large discrepancy reflects real behavior: the model is in a sensitive regime where small parameter changes cause qualitative changes in dynamics. In this case, the adjoint and FD are both locally valid; the discrepancy is genuine and should be reported as a finding (the model is near a threshold for these parameters).

Instructor note: This exercise mirrors real-world challenges in gradient-based optimization of climate models. The root cause (adaptive branching in reverse-mode AD) is well-documented in the scientific computing literature. Students who identify only "there is a bug" without specifying the mechanism receive partial credit. The key insight is that AD through adaptive-step ODE solvers is non-trivial, and the discrete adjoint (Chapter 6's approach) is the standard remedy.

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8 Chapter 8: Caveat Emptor

Exercise 8.1 *[Bloom L3]*

Solution.

$g(u;p) = u^3 - 3pu + 2 = 0$ (same as Ex 4.2, now with bifurcation focus).

(a) Real roots at $p = 1$ and most rapidly changing. Roots: $u_1 = 1$ (double), $u_2 = -2$. The double root $u_1 = 1$ changes most rapidly — its sensitivity is undefined (infinite), as shown in Ex 4.2. It is at a fold bifurcation point.

(b) $S_p^{u^*}$ at each root. As computed in Ex 4.2: $S_p^{u_1}$ is undefined (singular FSE); $S_p^{u_2} = 1/3$.

(c) Root curve and bifurcation. The real roots satisfy $u^3 - 3pu + 2 = 0$. By Cardano's formula or numerical continuation, the two positive roots merge at the fold bifurcation where $\partial g/\partial u = 3u^2 - 3p = 0$, i.e., $u = \sqrt{p}$. Substituting: $p^{3/2} - 3p\sqrt{p} + 2 = 0 \Rightarrow -2p^{3/2} + 2 = 0 \Rightarrow p^* = 1$. The two roots $u_1 = 1$ and $u_2 = -2$ merge at $p = 1$ (actually only the double root merges; $u_2 = -2$ persists). Careful: the cubic has the double root at $u = 1$ for $p = 1$. For $p < 1$, the double root splits into two complex roots (no real positive root near 1), and for $p > 1$, two distinct real roots appear near $u = 1$. The bifurcation is at $\mathbf{p = 1}$.

(d) FSE prediction at bifurcation. At $p = 1$, $\partial g/\partial u = 0$: the FSE $(\partial g/\partial u)(\partial u/\partial p) = -\partial g/\partial p$ has a zero coefficient, so $\partial u/\partial p = \infty$. Finite-difference verification: $u(p = 1.001)$ vs $u(p = 0.999)$ shows very large differences near the double root, confirming infinite sensitivity. The FD estimate diverges as $p \to 1$, consistent with a fold bifurcation.

Instructor note: This exercise directly connects Chapter 4 (implicit FSE) to Chapter 8 (bifurcations). The key point: the FSE singularity *is* the bifurcation. Students who answer only "sensitivity is large" without connecting to the eigenvalue of the Jacobian receive partial credit.

Exercise 8.2 *[Bloom L3]*

Solution.

(a) $S_\beta^{I(T)}$ at four values, $T = 90$ days.

Computed via centered FD with $h = 6 \times 10^{-6}\beta^*$:

β	$R_0 = k\beta\tau$	$I(90)$	$S_\beta^{I(90)}$ (approx.)
0.04	1.40	~ 150	$\sim +3.2$
0.06	2.10	~ 480	$\sim +1.8$
0.08	2.80	~ 620	$\sim +1.2$
0.12	4.20	~ 720	$\sim +0.7$

The SI *decreases* as R_0 increases past 1: at higher R_0, almost everyone has been infected by day 90, and $I(90)$ is in the declining tail, less sensitive to β.

(b) $\beta = 0.02$ ($R_0 = 0.7 < 1$). No epidemic occurs: $I(90) \approx I(0) = 1$ (infection dies out exponentially). $S_\beta^{I(90)}$ is small and positive near zero. **Qualitative change:** the SI is negligibly small for $R_0 < 1$ because no epidemic amplification occurs, but large and positive for $R_0 > 1$. Near $R_0 = 1$ ($\beta \approx 0.029$), the SI would be extremely large — the threshold crossing amplifies any perturbation enormously.

(c) Bifurcation diagram. The endemic equilibrium $\bar{I}^*$ bifurcates from $\bar{I}^* = 0$ at $R_0 = 1$ (transcritical bifurcation). For $R_0 < 1$: $\bar{I}^* = 0$ (disease-free equilibrium is stable). For $R_0 > 1$: $\bar{I}^* > 0$ (endemic equilibrium exists and is stable). The bifurcation diagram shows $\bar{I}^*$ vs. β as a forward (supercritical) transcritical bifurcation.

Instructor note: The key finding from part (a) is that $S_\beta^{I(T)}$ *decreases* as β increases — counterintuitive if one expected larger β to mean larger sensitivity. The reason: at high β, the epidemic has already run its course by day 90, and $I(90) \approx 0$ regardless of small perturbations to β. This illustrates why SA at a single time point can be misleading (Chapter 4, Ex 4.8).

Exercise 8.3 *[Bloom L4]*

Solution.

(a) Condition number and error vs. ϵ.

MATLAB code:

```
u_exact = [1;1;1];
eps_vals = [1, 0.1, 0.01, 0.001, 0.0001];
for k = 1:length(eps_vals)
    ep = eps_vals(k);
    A = [1 2 3; 4 5 6; 7 8 9+ep];
    b = A * u_exact;
    kappa(k) = cond(A);
    err(k) = norm(A\b - u_exact);
end
```

Expected results (approximate):

ϵ	$\kappa(A)$	$\|\hat{\vec{u}} - \vec{u}\|$
1.0	~ 30	$\sim 10^{-14}$
0.1	~ 300	$\sim 10^{-13}$
0.01	~ 3000	$\sim 10^{-12}$
0.001	$\sim 3 \times 10^4$	$\sim 10^{-11}$
0.0001	$\sim 3 \times 10^5$	$\sim 10^{-10}$

(b) Log-log slope. The relationship is $\|\hat{\vec{u}} - \vec{u}\| \propto \kappa(A)\, \varepsilon_{\text{mach}}$. On log-log axes, the slope is $+1$ (error grows linearly with condition number).

(c) Implication for large SI values.

A large SI for a nearly singular model reflects the ill-conditioning of the sensitivity computation, not necessarily a genuine large sensitivity of the physical model. If the Jacobian J_F of the model equations is nearly singular (large $\kappa(J_F)$), the FSE $J_F\,(\partial\vec{u}/\partial p_j) = \vec{f}_j$ will amplify any numerical errors in $\vec{f}_j$ by a factor of $\kappa(J_F)$. The reported SI may be $\kappa(J_F)$ times larger than the true value. *Practical implication:* always report $\kappa(J_F)$ alongside large SI values. An SI of 50 with $\kappa(J_F) = 100$ should be flagged as potentially unreliable; an SI of 50 with $\kappa(J_F) = 2$ is trustworthy.

Instructor note: The slope of 1 in part (b) is the classical condition-number error bound: $\|\delta\vec{u}\|/\|\vec{u}\| \leq \kappa(A)\|\delta\vec{b}\|/\|\vec{b}\|$. Students who compute the slope empirically and report it as ≈ 1 without citing the bound receive full credit; students who derive the bound analytically receive bonus credit.

Exercise 8.4 *[Bloom L5]*

Solution.

Advisory report: SA near the epidemic threshold ($R_0 = 1.04$).

(a) What local SA can and cannot tell you.

Local SA provides the derivative of each QOIwith respect to each parameter at the current best-fit values. With $R_0 = 1.04$, the model is very close to the epidemic threshold. Local SA *can* tell you which parameters would, if perturbed slightly, have the largest marginal effect on disease burden. What it *cannot* tell you: whether a parameter change of realistic magnitude (say 10–20%) would push R_0 below 1 (eliminating the epidemic entirely) or keep it above 1. Near the threshold, the model's behavior changes qualitatively, and local SA captures only the linearized behavior in an infinitesimally small neighborhood.

(b) Specific numerical problems near $R_0 = 1$.

Near the transcritical bifurcation at $R_0 = 1$, the Jacobian of the model at the endemic equilibrium has an eigenvalue near zero. This causes: (1) the FSE linear system $J_F\,(\partial\vec{u}/\partial p) = \vec{f}$ to be nearly singular, amplifying roundoff errors by $\kappa(J_F)$; (2) time-dependent SIs to grow without bound near the threshold crossing; (3) the adjoint ODE to have a slowly decaying mode (near-zero eigenvalue), making backward integration unstable over long time horizons. Practitioners should report $\kappa(J_F)$ and check that SI values are stable to perturbations of the nominal parameter set within the confidence region.

(c) Additional analyses recommended.

1. *Bifurcation diagram:* Plot the endemic equilibrium I^* vs. R_0 (or vs. the most controllable parameter). Determine how large a reduction in β or k is needed to push R_0 below 1. This gives a specific, actionable target rather than a marginal SI value.
2. *SA at multiple R_0 values:* Compute SI at $R_0 = 0.9$, 1.0, 1.04, 1.2. Report how the rankings change. If the ranking is stable across this range, the local SA at $R_0 = 1.04$ is informative. If not, flag the instability.
3. *Finite-change analysis:* Compute the actual change in peak I for a 10% reduction in each parameter. Report these directly alongside the SI values. For parameters near the threshold, the finite change may be qualitatively different from the SI prediction.

(d) Explaining bifurcation to a policy maker.

Think of the epidemic threshold as a tipping point. Below it ($R_0 < 1$), each infected person infects on average less than one other person, and the outbreak fades on its own. Above it ($R_0 > 1$), each infected person infects more than one other person, and the outbreak grows into an epidemic. At $R_0 = 1.04$, we are just above the tipping point. Even a small reduction in contact rates or transmission probability — say 5% — could push us below the tipping point, causing the epidemic to die out rather than sustain itself. This is the most important insight from the sensitivity analysis: at R_0 close to 1, small interventions can have disproportionately large effects. But it also means our predictions are highly uncertain: if the true R_0 is 0.96 rather than 1.04, the epidemic might already be declining naturally.

Instructor note: This is a realistic scenario that every applied mathematical modeler faces. The report format requirement is deliberate: public health advisors need concise, clear guidance, not a technical treatise. Deduct credit for responses that are technically correct but not accessible to a non-mathematical reader. The bifurcation explanation (part d) should use no equations.

Exercise 8.5 *[Bloom L6]*

Solution.

Three methodological concerns with the malaria SA ($R_0 = 1.15$, all SI < 0.1):

Concern 1: Small SIs near $R_0 = 1$ may be numerical artifacts.

Problem: With $R_0 = 1.15$, the model is moderately above the threshold, but close enough that the endemic equilibrium Jacobian may be ill-conditioned. Small SI values could result from condition-number amplification rather than genuine insensitivity.

Evidence needed: Report $\kappa(J_F)$ at the endemic equilibrium and verify that $|\mathrm{SI}| \gg \kappa(J_F)\,\varepsilon_{\mathrm{mach}}$ for each parameter. If any SI is within a factor of $\kappa(J_F)$ of machine epsilon, that SI is numerically unreliable.

Alternative analysis: Compute SIs of R_0 itself (analytically) and compare sign and magnitude with the equilibrium SIs. If $S^{R_0}_{p_j}$ is large but $S^{\bar{I}}_{p_j}$ is small, the discrepancy suggests numerical issues at the equilibrium.

Concern 2: SA of the endemic equilibrium does not address transient dynamics or the probability of outbreak onset.

Problem: If malaria is currently at or near the endemic equilibrium, the relevant policy question is "which intervention would most reduce endemic prevalence?" But if the model is used for outbreak forecasting (a different but common use), the relevant QOI is the peak or cumulative burden during a transient following an introduction event. SA of the endemic equilibrium cannot answer the latter question.

Evidence needed: Clarify the model's purpose. If forecasting an outbreak, compute SA for the transient QOI (e.g., peak I at $T = 180$ days) in addition to the endemic equilibrium.

Alternative analysis: Compute time-dependent SIs $S^{I}_{p_j}(t)$ and report whether the parameter ranking changes between the transient and endemic phases.

Concern 3: "All indices less than 0.1" does not imply robustness unless the uncertainty ranges are accounted for.

Problem: A local SI of 0.1 means a 1% change in the parameter produces a 0.1% change in prevalence. But if a parameter is uncertain by $\pm 50\%$ (plausible for entomological parameters in malaria models), the expected change in prevalence from this parameter alone is $\approx 5\%$. Reporting only SI values without the uncertainty range gives an incomplete picture.

Evidence needed: Report sigma-normalized SIs or one-way sensitivity ranges: the change in $\bar{I}$ when each parameter varies over its full uncertainty range. Parameters with small SI but large uncertainty range may still dominate the output variance.

Alternative analysis: Compute Sobol first-order and total indices across the plausible parameter uncertainty ranges. A parameter with local SI $= 0.1$ and uncertainty range $\pm 80\%$ may have Sobol $S_1 = 0.3$.

Instructor note: Full credit requires all three concerns with (a), (b), (c) for each. The three concerns should be genuinely distinct: numerical reliability (Concern 1), scope of the

SA (Concern 2), and accounting for uncertainty ranges (Concern 3). Students who only note “local SA doesn’t capture nonlinearity” receive partial credit for Concern 3 without the sigma-normalized index framing. The claim “all indices less than 0.1” is particularly dangerous near a threshold — this is the chapter’s core message.

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9 Chapter 9: Global Sensitivity Analysis

Exercise 9.1 *[Bloom L3]*

Solution.

Given: $S^{R_0}_{\tau_E} = 0.6$, $S^{R_0}_{\tau_I} = 1.2$, $S^{R_0}_{\beta} = 1.0$, $S^{R_0}_{k} = 1.0$, nominal $R_0^* = 2.0$. Uncertainties: $\sigma_{\tau_E} = 2$ d, $\sigma_{\tau_I} = 3$ d, $\sigma_\beta = 0.01$, $\sigma_k = 0.8$. Nominal values: $\tau_E^* = 5.5$ d, $\tau_I^* = 7$ d, $\beta^* = 0.06$, $k^* = 5$.

(a) Output standard deviation (delta method).

$\sigma^2_{R_0} = \sum_j \left(\frac{\partial R_0}{\partial p_j}\right)^2 \sigma^2_{p_j} = \sum_j \left(\frac{R_0^*}{p_j^*} S^{R_0}_{p_j}\right)^2 \sigma^2_{p_j}$.

$$\left(\frac{R_0^*}{\tau_E^*} S^{R_0}_{\tau_E}\right)^2 \sigma^2_{\tau_E} = \left(\frac{2}{5.5} \times 0.6\right)^2 \times 4 = (0.218)^2 \times 4 = 0.190$$
$$\left(\frac{R_0^*}{\tau_I^*} S^{R_0}_{\tau_I}\right)^2 \sigma^2_{\tau_I} = \left(\frac{2}{7} \times 1.2\right)^2 \times 9 = (0.343)^2 \times 9 = 1.059$$
$$\left(\frac{R_0^*}{\beta^*} S^{R_0}_{\beta}\right)^2 \sigma^2_{\beta} = \left(\frac{2}{0.06} \times 1.0\right)^2 \times (0.01)^2 = (33.3)^2 \times 10^{-4} = 0.111$$
$$\left(\frac{R_0^*}{k^*} S^{R_0}_{k}\right)^2 \sigma^2_{k} = \left(\frac{2}{5} \times 1.0\right)^2 \times (0.8)^2 = (0.4)^2 \times 0.64 = 0.102$$

$\sigma^2_{R_0} = 0.190 + 1.059 + 0.111 + 0.102 = 1.462$, $\sigma_{R_0} \approx \mathbf{1.21}$.

(b) Sigma-normalized indices.

$\tilde{S}_{p_j} = (R_0^*/p_j^*)\,|S^{R_0}_{p_j}|\,\sigma_{p_j}/\sigma_{R_0}$:

Parameter	$\tilde{S}$ (sigma-norm.)	Contribution (%)
τ_I	$(2/7)(1.2)(3)/1.21 = 0.851$	58%
τ_E	$(2/5.5)(0.6)(2)/1.21 = 0.360$	13%
β	$(2/0.06)(1.0)(0.01)/1.21 = 0.276$	8%
k	$(2/5)(1.0)(0.8)/1.21 = 0.264$	7%

(c) Tornado plot. Sorted by $\tilde{S}$: $\tau_I > \tau_E > \beta \approx k$.

(d) Ranking comparison. Local SI ranking: τ_I and k tied at 1.2 and 1.0; β at 1.0; τ_E at 0.6. Sigma-normalized ranking: $\tau_I \gg \tau_E > \beta \approx k$.

τ_I remains dominant in both, but its lead is much larger when uncertainty is accounted for ($\sigma_{\tau_I} = 3$ d is large). k drops from second to last: despite SI $= 1.0$, its uncertainty $\sigma_k = 0.8$ relative to $k^* = 5$ gives a coefficient of variation of only 16%, while τ_E has $\sigma_{\tau_E}/\tau_E^* = 36\%$. k **changed rank most dramatically** because it has a large SI but small relative uncertainty.

Instructor note: The sigma-normalized index is the key concept here: it combines the sensitivity (SI) with the actual uncertainty range to produce a practical priority ranking. Students who compute only SIs without accounting for uncertainty miss the point of this exercise.

Exercise 9.2 *[Bloom L3]*

Solution.

(a) LHS implementation and comparison.

```
rng(42);
N = 100; m = 3;
X_lhs = lhs_sample(N, m, zeros(1,m), ones(1,m));
X_rnd = rand(N, m);
```

(b) Scatter plots. The LHS scatter plots show more uniform coverage with fewer clusters; the random sample shows visible gaps and clusters in all three pairwise projections.

(c) Mean and standard deviation. Both LHS and random sampling should give means near 0.5 and standard deviations near $1/\sqrt{12} \approx 0.289$ for a uniform distribution on $[0, 1]$. LHS is typically closer to both exact values because the stratification ensures coverage of the full range. For $N = 100$, the improvement in mean accuracy is ∼30–50% and in standard deviation ∼20–40%.

(d) Histogram comparison. The LHS histogram shows a much flatter, more uniform distribution (each of the 100 strata has exactly one sample). The random sample histogram shows the typical Poisson variation (some bins have 0 or 3+ samples). For sensitivity analysis, the LHS coverage is preferable because it ensures the model is evaluated in every region of the parameter space.

Instructor note: Part (c) is the quantitative check: students should compute the actual mean and std and report how close they are to the theoretical values. The LHS improvement in mean accuracy comes from the fact that the sample mean of a stratified sample is an unbiased estimator with lower variance than the sample mean of a random sample (stratification theorem).

Exercise 9.3 *[Bloom L3]*

Solution.

(a-b) LHS + PRCC computation.

Using $N = 200$, $\pm 50\%$ around nominal values:

```
p = sir_nominal();
lb = 0.5*[p.k, p.beta, p.tau, p.L];
ub = 1.5*[p.k, p.beta, p.tau, p.L];
rng(42); X = lhs_sample(200, 4, lb, ub);
Y = arrayfun(@(i) sir_I_max(X(i,:), p), 1:200)';
[rho, pval] = compute_prcc(X, Y, {'k','beta','tau','L'});
```

Expected PRCC values (approximate):

Parameter	PRCC	p-value
k	$+0.85$	< 0.001
β	$+0.84$	< 0.001
τ	$+0.72$	< 0.001
L	-0.04	> 0.5

(c) PRCC bar chart. Sorted: $k \approx \beta > \tau \gg L$ (similar to SI ranking).

(d) Strongest negative PRCC: L. L (mean lifespan) has a negative PRCC: longer lifespans reduce peak infections. In plain language: populations with longer lifespans have smaller background mortality rates, which means the susceptible pool is replenished more slowly, and a larger fraction of the population is already immune/recovered by the time of peak infections. The effect is very small ($|\text{PRCC}| \approx 0.04$, not statistically significant).

(e) Comparison with local SI tornado plot. Both methods rank k and β highest, τ third, L negligible. The PRCC ranking agrees with the local SI ranking for this model. Agreement is expected because peak infections $I_{\max}$ is monotone in each parameter over the $\pm 50\%$ uncertainty range (the model is not near any threshold), satisfying the PRCC assumption.

Instructor note: For the SIR model with $R_0 = 2.1$ and $\pm 50\%$ variation, the PRCC and local SI rankings should agree well. Discrepancies would appear if: (a) β variation includes $R_0 < 1$ (non-monotone at threshold), or (b) parameter interactions are strong (nonlinear effects). Students should note that agreement here is specific to this model and range.

Exercise 9.4 *[Bloom L4]*

Solution.

$Y = 2Q_1^2 + Q_2$, $Q_1, Q_2 \sim \mathcal{U}(0,1)$, independent.

(a) Variance decomposition. $\mathbb{E}[Y] = 2\mathbb{E}[Q_1^2] + \mathbb{E}[Q_2] = 2/3 + 1/2 = 7/6$. $\operatorname{Var}(Y) = 4\operatorname{Var}(Q_1^2) + \operatorname{Var}(Q_2)$. $\operatorname{Var}(Q_1^2) = \mathbb{E}[Q_1^4] - (\mathbb{E}[Q_1^2])^2 = 1/5 - 1/9 = 4/45$. $\operatorname{Var}(Q_2) = 1/12$. $\operatorname{Var}(Y) = 4(4/45) + 1/12 = 16/45 + 1/12 = 64/180 + 15/180 = 79/180$.

$D_1 = \operatorname{Var}[\mathbb{E}(Y|Q_1)] = \operatorname{Var}[2Q_1^2 + 1/2] = 4\operatorname{Var}(Q_1^2) = 16/45$. $D_2 = \operatorname{Var}[\mathbb{E}(Y|Q_2)] = \operatorname{Var}[2/3 + Q_2] = 1/12$.

(b) Sobol indices. $S_1 = D_1/\operatorname{Var}(Y) = (16/45)/(79/180) = (16/45)(180/79) = 64/79 \approx 0.810$. $S_2 = D_2/\operatorname{Var}(Y) = (1/12)/(79/180) = (180/12)/79 = 15/79 \approx 0.190$. $S_1 + S_2 = 79/79 = 1.0$. Yes, they sum to 1, because the model is *additive* (no interaction terms), so the full variance is explained by the first-order indices.

(c) Total indices. For an additive model: $S_{T_1} = S_1 = 64/79$, $S_{T_2} = S_2 = 15/79$. Total indices equal first-order indices when there are no interactions.

(d) Adding interaction Q_1Q_2: $Y' = 2Q_1^2 + Q_2 + Q_1Q_2$. Without computing: adding Q_1Q_2 introduces an interaction term. S_1 decreases (some variance is now shared with Q_2); S_2 decreases for the same reason. The total indices $S_{T_1} > S_1$ and $S_{T_2} > S_2$ because each parameter contributes to the interaction term.

Verification: $\operatorname{Var}(Y') = 4\operatorname{Var}(Q_1^2) + \operatorname{Var}(Q_2) + \operatorname{Var}(Q_1Q_2) + 2\operatorname{Cov}(\ldots)$ (with Cov terms vanishing by independence). $\operatorname{Var}(Q_1Q_2) = \mathbb{E}[Q_1^2Q_2^2] - (\mathbb{E}[Q_1Q_2])^2 = (1/3)(1/3) - (1/2)^2(1/4) = 1/9 - 1/16 = 7/144$. $D_{12} = \operatorname{Var}(Y') - D_1' - D_2' -$ (remaining interaction variance). Computing exactly: $S_1' < 0.810$, $S_2' < 0.190$, $S_1' + S_2' < 1$. The interaction term $D_{12} > 0$ accounts for the remainder.

Instructor note: Part (b)'s exact sum-to-1 for the additive model is a crucial result: for any purely additive model, all first-order indices sum to 1 and total indices equal first-order indices. Part (d) demonstrates the effect of interactions without requiring exact computation, testing conceptual understanding over algebraic facility.

Exercise 9.5 *[Bloom L4]*

Solution.

Using `run_global_sa_sir` from the repository with $N = 2000$.

(a) Sobol indices (nominal parameters, $\pm 50\%$ range).

Approximate values from Jansen estimators:

Param	$\hat{S}_i$	$\hat{S}_{T_i}$	$\hat{S}_{T_i} - \hat{S}_i$
k	0.38	0.43	0.05
β	0.38	0.43	0.05
τ	0.20	0.25	0.05
L	0.00	0.00	0.00

(b) Parameters with $S_{T_i} \gg S_i$. For all three active parameters, $S_{T_i} - S_i \approx 0.05$, indicating moderate interaction effects (primarily k-β and k-τ interactions). No parameter has $S_{T_i} \gg S_i$ for this model at this range; interactions are present but modest. This is consistent with the SIR model being a smooth, moderately nonlinear ODE.

(c) Sums. $\sum S_i \approx 0.96 < 1$ (interaction terms absorb the remaining $\sim 4\%$). $\sum S_{T_i} \approx 1.11 > 1$ (interactions are double-counted in total indices). For a purely additive model, both sums would equal 1. The observed values confirm the presence of small but nonzero interactions.

(d) Best parameter to measure precisely. To reduce $\mathrm{Var}(I_{\max})$ most, measure the parameter with the largest S_{T_i} (total index), since S_{T_i} measures the variance that would remain if all uncertainty in parameter i were eliminated, including interactions. Here k and β are tied at $S_{T_i} \approx 0.43$. Measuring either k or β precisely reduces $\mathrm{Var}(I_{\max})$ by $\approx 43\%$ (more precisely, the variance conditional on knowing k exactly is $(1 - S_{T_k})\,\mathrm{Var}(I_{\max}) \approx 0.57\ \mathrm{Var}(I_{\max})$).

Instructor note: Part (d) is a key policy application of total indices: the total index, not the first-order index, answers "which parameter measurement would most reduce output uncertainty?" First-order indices answer "which parameter contributes most to variance independently?" Both are important, and students should be able to distinguish them.

Exercise 9.6 *[Bloom L4]*

Solution.

Using $r = 15$ Morris trajectories, $p = 4$ levels ($\Delta = 4/6 = 2/3$), $\pm 50\%$ parameter ranges.

(a) Morris measures. Approximate values from `morris_screening`:

Param	μ^*	σ	Classification
k	~ 80	~ 20	Important, nonlinear
β	~ 78	~ 19	Important, nonlinear
τ	~ 42	~ 14	Important, nonlinear
L	~ 1	~ 0.5	Negligible

(b) Diagnostic plot. The μ^* vs. σ plot shows k and β in the upper-right (large μ^* and σ: important and nonlinear), τ in the middle-right, and L near the origin (negligible).

(c) Most and least influential. Most: k (or β, tied). Least: L.

(d) Reduced Sobol (3 parameters, L fixed).

With L fixed: Sobol cost = $N(m+2) = 2000 \times 5 = 10{,}000$ evaluations vs. full $2000 \times 6 = 12{,}000$. Cost reduction: 17%. Approximate results: $S_k \approx 0.43$, $S_\beta \approx 0.43$, $S_\tau \approx 0.24$; total indices similar. Fixing L does not materially change the other indices (since L was negligible). The Morris screening correctly identified that L could be dropped.

Instructor note: The Morris cost here is $15 \times (4+1) = 75$ evaluations vs. 12,000 for full Sobol — a 160× cost reduction for the screening phase. This illustrates Morris's value as a cheap first step that identifies which parameters can be dropped from the expensive Sobol analysis.

Exercise 9.7 *[Bloom L5]*

Solution.

This exercise requires the student to choose and describe a specific published model. The model answer below uses the Chitnis dengue model [?] as an example; any 8+ parameter epidemic model with published parameter estimates is acceptable.

(a) Model, parameters, uncertainty ranges. Chitnis et al. (2008) dengue model: 8 state variables, 14 parameters including mosquito biting rate b_v, transmission probabilities β_{vh} and β_{hv}, mosquito and human mortality rates, and mean durations. Uncertainty ranges: ±50% for biological parameters (based on reported ranges in Table 2 of the paper); ±30% for demographic parameters (population data are more precisely known).

(b) QOI justification. Primary: R_0 (determines whether dengue can persist in the population — the most policy-relevant threshold). Secondary: equilibrium human infection prevalence $\bar{I}_h/N_h$ (determines endemic burden for healthcare planning).

(c) PRCC vs. Sobol choice. Use **Sobol indices**. The dengue model is moderately nonlinear and has known interaction between biting rate b_v and transmission probabilities ($R_0 \propto b_v^2 \beta_{vh}\beta_{hv}$, so interactions are structural). PRCC assumes monotone relationships; near threshold or at high R_0, this assumption may not hold for all parameters. Sobol indices capture interactions and do not require monotonicity.

(d) Sample size N. Use $N = 2000$ for each Sobol matrix. Run the convergence check: compute Sobol indices at $N = 200, 500, 1000, 2000$; report the value of N at which the first-order indices change by $< 2\%$ as N doubles. For a 14-parameter model, the total evaluation cost is $N(14+2) = 32{,}000$ runs. If each run takes 2 minutes, this is ≈ 4.4 days of computation on one core.

(e) Presentation format. Two tornado plots of Sobol S_{T_i} (one per QOI), with parameters labeled as interventions: "Reducing mosquito biting rate by X% through bed nets reduces R_0 by approximately Y%." A two-panel figure showing which parameters appear in both tornado plots (co-optimization) and which are QOI-specific. Use traffic-light color coding: red for high-priority, green for low-priority intervention targets.

(f) Potential misleading results. If the model's best-fit parameters give R_0 near 1 (endemic/non-endemic threshold), Sobol indices at those nominal values will be dominated by parameters near the threshold. A 2% uncertainty in β_{vh} might shift the model from endemic to non-endemic, giving extremely large apparent sensitivity for β_{vh}. Recommendation: compute Sobol indices at three nominal parameter sets ($R_0 = 1.5$, 2.0, 3.0) and report whether the parameter rankings are stable across this range.

Instructor note: This is an open-ended design exercise. Full credit requires all six components with operationally specific content. Vague statements ("choose the right sample size," "present clear figures") without specifics receive partial credit. The model choice affects all answers; a reasonable choice with internally consistent answers earns full credit even if different from the example above.

Exercise 9.8 *[Bloom L6]*

Solution.

Methodological concerns with Sobol analysis ignoring correlation among three parameters.

(a) Mathematical problem.

Standard Sobol indices assume parameter *independence*: the matrices A and B are sampled independently from the joint distribution $p(\vec{p}) = \prod_j p_j(p_j)$. When three parameters are correlated, the correct joint distribution is $p(\vec{p}) = p(p_1, p_2, p_3) \times \prod_{j>3} p_j(p_j)$, which cannot be factored. Sampling independently from marginal distributions and then compositing (the Saltelli method) generates non-physical parameter combinations (e.g., high phosphorus loading but low nitrogen loading, which does not occur in the real watershed system). The resulting Sobol indices estimate sensitivity to artificially independent variations, not to the actual coupled variations that the system experiences.

(b) Effect on each correlated parameter's index.

Phosphorus loading P: The standard $S_1(P)$ likely *overestimates* the true first-order index. When P and N are positively correlated, the actual variation of P is accompanied by variation in N. The standard estimator attributes only the P variation to $S_1(P)$; the interaction that would occur under the true distribution is ignored, artificially inflating the apparent individual contribution of P.

Nitrogen loading N: Same logic applies: $S_1(N)$ likely overestimates.

Water residence time T_w: If T_w is correlated with both P and N (longer residence = more loading), $S_1(T_w)$ may be *underestimated* because the positive correlation means that varying T_w while holding P and N fixed understates the total effect of residence time in the real system.

(c) Minimal correction.

Use a copula-based sampling method that preserves the observed correlations. The Iman–Conover rank correlation constraint method is the simplest: (1) generate independent LHS samples; (2) apply the Iman–Conover algorithm to induce the observed rank correlations among the three parameters; (3) compute Sobol indices using the correlated sample matrices. This adds one matrix transformation step to the existing workflow with no additional model evaluations.

(d) Why sum-to-1 is not a valid independence check.

The Sobol first-order indices sum to 1 only for an additive model with independent parameters. With correlated parameters, the indices can still sum to approximately 1 even with incorrect independence assumptions, because the sum reflects the total explained variance, which equals 1 by construction (the variance is fully accounted for by some combination of first-order and interaction terms). The sum being 1 says nothing about whether the individual index attributions are correct — it is a property of the decomposition algorithm, not of the physical model or the sampling design.

Instructor note: This is a rigorous L6 exercise testing critical evaluation of a real methodological error. Part (d) is the subtlest: students often think "indices sum to 1" is a validity check, but it is not — it is simply a consequence of the ANOVA decomposition. Full credit requires a clear mathematical explanation, not just "the check doesn't work."
